

1. Questions

Read the following information carefully and answer the questions based on it.

The table chart given below shows the percentage of the number of veg rolls made out of the total number of rolls (veg + non-veg) made by four different shops A, B, C and D. It also shows the ratio of the number of veg and non-veg rolls sold and some information in the form of statement.

Shops	The percentage of the number of veg rolls made	The ratio of the number of veg and non-veg rolls sold	Statement
A	40%	9:16	The difference between the number of veg and non-veg rolls unsold is 20 (veg > non-veg)
B	60%	5:3	The number of veg rolls unsold is equal to the number of non-veg rolls unsold
C	50%	4:3	The number of non-veg rolls unsold is half of the number of veg rolls sold
D	80%	6:1	The number of veg rolls unsold is $\frac{2}{5}$ th of the number of non-veg rolls sold

Note:

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- i). The total number of rolls made = the number of veg rolls made + the number of non-veg rolls made
- ii). The total number of veg rolls made = the number of veg rolls sold + the number of veg rolls unsold
- iii). The total number of non-veg rolls made = the number of non-veg rolls sold + the number of non-veg rolls unsold
- iv). The total number of rolls (veg + non-veg) made by shop A is 600.
- v). The total number of non-veg rolls made by shop B is 44.44% less than shop A.
- vi). The ratio of the number of veg rolls made by shop B and C is 6:7.

vii). The total number of rolls (veg + non-veg) made by shop D is 100 more than shop C.

In shop A, the price of each non-veg roll is 50% more than the price the of each veg roll, while the total revenue generated from the sale of all rolls (veg + non-veg) is Rs. 66000.

Which of the following can be determined?

I). The difference between the price of each veg and non-veg roll sold by shop A.

II). The total revenue generated by shop A from the sale of all the veg rolls.

III). The percentage by which the total revenue generated by shop A from the sale of all veg rolls is more or less than the total revenue generated by the same shop from the sale of all non-veg rolls.

- a. Only I
- b. Both I and II
- c. Both II and III
- d. All I, II and III
- e. None can be determined

2. Questions

Out of the total number of veg rolls sold by shop B, some are sold to adults (males + females), while the rest are sold to children (boys + girls). If the ratio of the number of veg rolls sold to females and girls is 4:1, while the number of veg rolls sold to boys is 30, which is $\frac{1}{4}^{\text{th}}$ of the number of veg rolls sold to males, then find the total number of veg rolls sold to adults (males + females) is what percent of the total number of veg rolls sold in shop B?

- a. 80%
- b. 75%
- c. 60%
- d. 55%
- e. None of these

3. Questions

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The total number of pizzas (veg + non-veg) sold by shop D is 14.28% less than the total number of rolls (veg + non-veg) sold by the same shop. If the price of each non-veg pizza in shop D is Rs. 240, which is 20% more than the price of each veg pizza in the same shop and the total revenue generated from the sale of all pizzas (veg + non-veg) by shop D is Rs. 126000, then find the total number of non-veg pizzas sold by shop D.

- a. 200
- b. 150
- c. 250

- d. 100
- e. None of these

4. Questions

There are three varieties of veg rolls are sold by shop C i.e., cheese, tomato and onion. If the ratio of the number of tomato and onion veg rolls sold is 9:7, while the number of cheese veg rolls sold is 30 more than the number of tomato veg rolls sold, then the number of onion veg rolls sold by shop C will be ____[?].

What should come in place of question mark?

- a. 63
- b. 56
- c. 49
- d. 70
- e. None of these

5. Questions

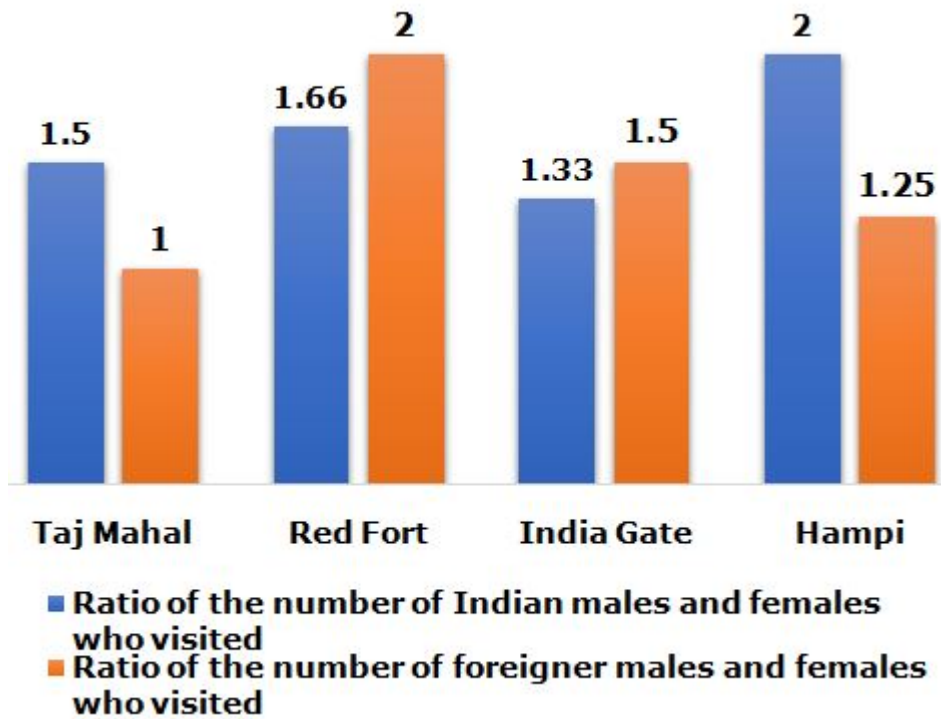
The total number of rolls (veg + non-veg) made by another shop E is equals to shop B, while the number of non-veg rolls unsold by shop E is 10, which is $\frac{1}{10}^{\text{th}}$ of the number of veg rolls unsold by the same shop. If the number of veg rolls sold is 210 more than the number of non-veg rolls sold by shop E, then find the number of veg rolls sold by shop E.

- a. 300
- b. 180
- c. 210
- d. Cannot be determined
- e. None of these

6. Questions

Read the following information carefully and answer the questions based on it.

The bar graph given below shows the ratio of the number of Indian males and females who visited four different places namely Taj Mahal, Red Fort, India Gate and Hampi. It also shows the ratio of the number of male and female foreigners who visited each place.



Notes:

- i). The total number of visitors who visited = the number of Indian visitors who visited + the number of foreigner visitors who visited
- ii). The total number of Indian visitors who visited = the number of Indian males who visited + the number of Indian females who visited
- iii). The total number of foreigners visitors who visited = the number of foreigners males who visited + the number of foreigners females who visited
- iv). The total number of visitors who visited Taj Mahal is 700, out of which $\frac{4}{7}^{\text{th}}$ are males (Indians + Foreigners).
- v). The total number of males (Indians + Foreigners) who visited Red Fort is 450, which is 80% more than the total number of females (Indians + Foreigners) who visited the same place.
- vi). The total number of visitors who visited India Gate is 500, which is 550 less than that of Hampi.
- vii). The total number of females (Indians + Foreigners) who visited Hampi is 400, which is 190 more than that of India Gate.

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Out of the total number of Indians (male + female) who have visited Taj Mahal, some are employed while the rest are unemployed. Of the total number of employed Indians, 80% are graduates and the rest are post-graduates. If the number of post-graduate Indians who visited the Taj Mahal is 20 less than the number of unemployed Indians who visited the same place, then find the ratio of the number of employed and unemployed Indians who visited the Taj Mahal.

- a. 4:1
- b. 3:2
- c. 5:3

d. Cannot be determined

e. None of these

7. Questions

If the number of foreigner females from UK who visited Red Fort is _____, which is 50% of the number of foreigner males from UK who visited the same place, then the total number of foreigner (males + females) from USA who visited Red Fort becomes _____. (The foreigners who visited Red Fort are from UK and USA only)

Which of the following does satisfy the filler(s) in the same order?

I). 90 and 30

II). 80 and 50

III). 70 and 80

a. Both II and III follow

b. Both I and II follow

c. Only I follow

d. All I, II and III follow

e. None follow

8. Questions

If the number of illiterate Indian males and females who visited India Gate is equal. If the number of literate Indian males who visited India Gate is 40% more than the number of literate Indian females who visited the same place, then find the total number of literate Indians (males + females) who visited India Gate.

a. 280

b. 300

c. 240

d. 360

e. None of these

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9. Questions

The total number of foreigners who visited another place Konark Sun Temple is 300, which is 60% of the total number of Indians who visited the same place. If the ratio of the number of Indian males and foreigner females who visited Konark Sun Temple is 3:1, while the number of Indian females and foreigner males are equal, then find what percent of the total number of Indians who visited Konark Sun Temple are males?

a. 50%

- b. 75%
- c. 60%
- d. 45%
- e. None of these

10. Questions

If the number of Indian males who visited Hampi is reduced by $[X]$ and the number of Indian females who visited Hampi is increased by $[2X]$, then the ratio of the number of Indian males and females who visited Hampi will be 7:6. The value of X is exactly divisible by $___[?]__$.

What should come in place of question mark?

- a. 12
- b. 18
- c. 15
- d. 10
- e. None of these

11. Questions

Read the following information carefully and answer the questions based on it.

Two shops X and Y sells two different types of watches i.e., digital watch and smart watch.

Shop X: The cost price of the digital watch and the selling price of the smart watch are the same. The marked price of the digital watch is Rs. 1800, which is sold at a discount of 11.11%. The smart watch is marked up by 40% over its cost price and sold at a profit of 20%. The total discount allowed on the sale of the digital watch and smart watch together is Rs. 400.

Shop Y: The cost price of the digital watch is Rs. 2000, which is sold at a profit of 50%. The selling price of the smart watch is Rs. 2500, which is sold at a discount of 16.67%. The ratio of the marked price of the digital watch to the cost price of the smart watch is 32:15. The amount by which the digital watch has been marked up is Rs. 300 less than the amount by which the smart watch has been marked up.

Shop Z: The marked price of digital watch is Rs. 840 more than the marked price of the smart watch. The digital watch has been marked up by 25% above its cost price and sold for Rs. 2700. The cost price of the smart watch is Rs. 1800, which is sold at a discount of Rs. 160. The ratio of the cost price of the digital watch and the selling price of the smart watch is 6:5.

Notes:

The total number of watches sold = the number of digital watches sold + the number of smart watches sold

The profit should be calculated on cost price, unless mentioned in the question.

In shop X, if there is an additional discount of Rs. _____ was allowed on selling the digital watch, then the profit earned on selling the digital watch in shop X will be _____%. (The C.P. and the M.P.

remained unchanged)

Which of the following does satisfy the filler(s) in the same order?

- I). 100 and 25
- II). 200 and 20
- III). 300 and 10
- a. Only I follow
- b. Both I and II follow
- c. Both II and III follow
- d. All I, II and III follow
- e. None follow

12. Questions

Out of the total number of smart watches purchased by Shop Y, the number of smart watches that remained unsold is 4, while the rest of the smart watches are sold. If the total profit earned by Shop Y on all the smart watches sold is Rs. 2000, then find the total number of smart watches purchased by Shop Y.

- a. 18
- b. 12
- c. 16
- d. 14
- e. None of these

13. Questions

In another shop W, the cost price of digital is Rs. 800, which is marked-up by $[M]\%$ above its cost price and sold at a discount of Rs. 112, then the profit earned on selling the digital watch is 26%. If the cost price of the smart watch in shop W is Rs. $[16M+60]$ and sold at a profit of 10%, then find the selling price of the smart watch in shop W.

- a. Rs. 600
- b. Rs. 560
- c. Rs. 480
- d. Rs. 770
- e. None of these

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14. Questions

The marked price of analog watch in shop Z is Rs. $[K]$, which is sold at a profit of Rs. 200, while the

selling price of the analog watch is Rs. 700. If the analog watch is marked-up by 60% above its cost price, then find the difference between the marked price of the analog and digital watch in shop Z.

- a. Rs. 2200
- b. Rs. 1500
- c. Rs. 3000
- d. Cannot be determined
- e. None of these

15. Questions

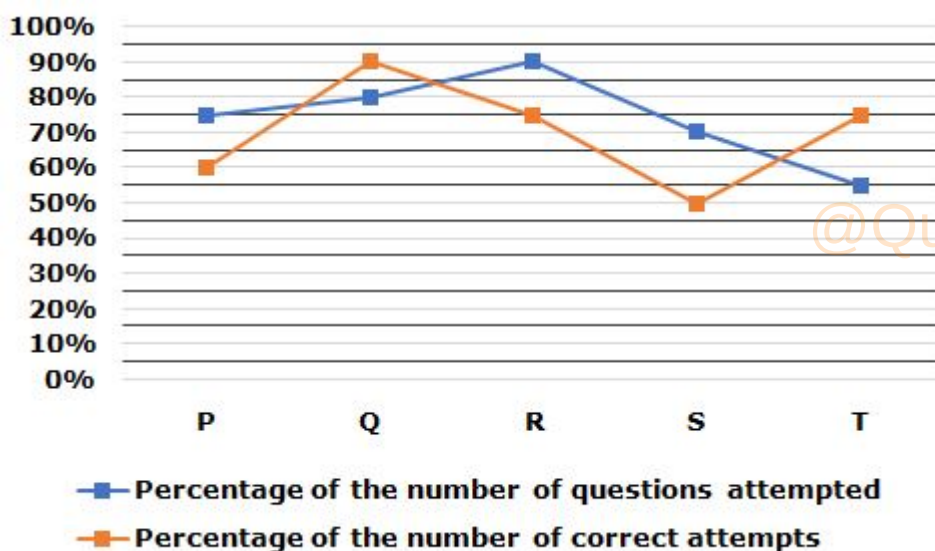
In shop Z, if the marked price of the smart watch is Rs. [L], while the cost price and selling price of the smart watch remained unchanged, the discount allowed on smart watch in shop Z will be Rs. 400. Find the value of $\sqrt{L+100}$.

- a. 40
- b. 20
- c. 50
- d. 30
- e. None of these

16. Questions

Read the following information carefully and answer the questions based on it.

The line graph given below shows the percentage of the number of questions attempted by a student in five different subjects P, Q, R, S and T. It also shows the percentage of the number of correct attempts made out of the total number of questions attempted (correct + wrong) by the student in each subject.



Notes:

- i). The total number of questions contained = the number of questions attempted + the number of questions skipped
- ii). The total number of questions attempted = the number of correct attempts + the number of wrong attempts
- iii). The ratio of the total number of questions contained in subject P and Q is 3:5.
- iv). The number of wrong attempts made in subject P is 20 more than subject Q.
- v). The total number of questions contained in subject R is half as that of subject S.
- vi). The number of correct attempts made in subject R is 2 less than subject S.
- vii). The number of questions skipped in subject T is 60 more than subject S.

In subject P, if the marks deducted for each wrong attempt is [A], which is $\frac{1}{4}^{\text{th}}$ of the marks awarded for each correct attempt, then the total marks obtained by the student in subject P will be 45.

Which of the following is/are definitely about [A]?

- I). Positive integer
- II). Whole number
- III). Prime number
- a. Only I
- b. Both I and II
- c. Both II and III
- d. All I, II and III
- e. None is true

17. Questions

Subject Q is divided into two parts i.e., subjective and objective. The marks awarded for each correct attempt made by the student in subjective section is 1, which is twice as that of objective section and the marks deducted for each wrong attempt in objective section is 0.10, which is $\frac{2}{5}^{\text{th}}$ as that of subjective section. If 8.33% and 25% of the total number of correct and the total number of wrong attempts respectively are from objective section, then find the total marks obtained by the student in subject Q.

- a. 142.80
- b. 134.60
- c. 128.20
- d. 150.50
- e. None of these

18. Questions

In subject S, the number of questions attempted in theory section is $__\text{[X]}__\$, which is 28 more than the number of questions skipped in theory section. If $\frac{1}{4}^{\text{th}}$ of the total number of questions skipped are from practical section, then the number of questions attempted in practical section will be $__\text{[Y]}__\$. (There are only two sections i.e., theory and practical)

- a. $X > Y$
- b. $X < Y$
- c. $X \geq Y$
- d. $X \leq Y$
- e. $X = Y$

19. Questions

If the ratio of the number of questions attempted and skipped by another student in subject R is 3:1, while the total marks obtained by another student in subject R is 55, then find what percentage of the total number of questions attempted by another student in subject R are correct? (1 mark awarded for each correct attempt and 0.25 mark deducted for each wrong attempt)

- a. 83.33%
- b. 66.67%
- c. 93.33%
- d. Cannot be determined
- e. None of these

20. Questions

The total number of questions contained in another subject U is 280, out of which the number of questions skipped is [2Y] , which is $\frac{2}{3}^{\text{rd}}$ of the number of wrong attempts in the same subject. Out of the total number of questions attempted in subject U, if 83.33% questions are correct, then find the difference between the total number of questions attempted by the student in subject T and U.

- a. 120
- b. 144
- c. 108
- d. 136
- e. None of these

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Explanations:

1. Questions

Shop A

The total number of rolls made = 600

The total number of veg rolls made = 40% of 600 = 240

The total number of non-veg rolls made = 60% of 600 = 360

Let the number of veg and non-veg rolls sold be $9n$ and $16n$ respectively.

So, the number of veg rolls unsold = $(240-9n)$

And, the number of non-veg rolls unsold = $(360-16n)$

Now,

$$(240-9n) - (360-16n) = 20$$

$$220 - 9n = 360 - 16n$$

$$7n = 140$$

$$n = 20$$

Therefore,

The number of veg rolls sold = $9n = (9 \times 20) = 180$

The number of non-veg rolls sold = $16n = (16 \times 20) = 320$

The number of veg rolls unsold = $(240-9n) = (240-180) = 60$

The number of non-veg rolls unsold = $(360-16n) = (360-320) = 40$

Shop B

The total number of non-veg rolls made = $5/9$ of 360 = 200 (Note: $44.44\% = 4/9$)

The total number of veg rolls made = $\{(200/40) \times 60\} = 300$

The total number of rolls made = $(300+200) = 500$

Let the number of veg and non-veg rolls sold be $5m$ and $3m$ respectively.

So, the number of veg rolls unsold = $(300-5m)$

And, the number of non-veg rolls unsold = $(200-3m)$

Now,

$$(300-5m) = (200-3m)$$

$$2m = 100$$

$$m = 50$$

Therefore,

The number of veg rolls sold = $5m = (5 \times 50) = 250$

The number of non-veg rolls sold = $3m = (3 \times 50) = 150$

The number of veg rolls unsold = $(300 - 5m) = (300 - 250) = 50$

The number of non-veg rolls unsold = $(200 - 3m) = (200 - 150) = 50$

Shop C

The number of veg rolls made = $\{(300/6) \times 7\} = 350$

The number of non-veg rolls made = $\{(350/50) \times 50\} = 350$

The number of total number of rolls made = $(350 + 350) = 700$

Let the number of veg and non-veg rolls sold be $4k$ and $3k$ respectively.

So, the number of non-veg rolls unsold = $(4k/2) = 2k$

And, the number of veg rolls unsold = $(350 - 4k)$

Now,

$$3k + 2k = 350$$

$$5k = 350$$

$$k = 70$$

Therefore,

The number of veg rolls sold = $4k = (4 \times 70) = 280$

The number of non-veg rolls sold = $3k = (3 \times 70) = 210$

The number of veg rolls unsold = $(350 - 4k) = (350 - 280) = 70$

The number of non-veg rolls unsold = $2k = (2 \times 70) = 140$

Shop D

The total number of rolls made = $(700 + 100) = 800$

The total number of veg rolls made = 80% of $800 = 640$

The total number of non-veg rolls made = 20% of $800 = 160$

Let the number of veg and non-veg rolls sold be $6j$ and j respectively.

So, the number of veg rolls unsold = $(640 - 6j)$

And, the number of non-veg rolls unsold = $(160 - j)$

Now,

$$(640 - 6j)/j = 2/5$$

$$3200 - 30j = 2j$$

$$32j = 3200$$

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$$j = 100$$

Therefore,

$$\text{The number of veg rolls sold} = 6j = (6 \times 100) = 600$$

$$\text{The number of non-veg rolls sold} = j = 100$$

$$\text{The number of veg rolls unsold} = (640 - 6j) = (640 - 600) = 40$$

$$\text{The number of non-veg rolls unsold} = (160 - j) = (160 - 100) = 60$$

Veg Rolls

Shop	The total number of veg rolls made	The number of veg rolls sold	The number of veg rolls unsold
A	240	180	60
B	300	250	50
C	350	280	70
D	640	600	40
Total			

Non-veg Rolls

Shop	The total number of non-veg rolls made	The number of non-veg rolls sold	The number of non-veg rolls unsold
A	360	320	40
B	200	150	50
C	350	210	140
D	160	100	60
Total			

Answer: D

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Now, according to the question,

Shop A

$$\text{The total number of veg rolls sold} = 180$$

$$\text{The total number of non-veg rolls sold} = 320$$

Let the price of each veg roll be Rs. 100p

$$\text{So, the price of each non-veg roll} = 150\% \text{ of } 100p = \text{Rs. } 150p$$

$$\text{Now, the total revenue generated from the sale of all veg rolls} = (180 \times 100p) = \text{Rs. } 18000p$$

And, the total revenue generated from the sale of all non-veg rolls = $(320 \times 150p) = \text{Rs. } 48000p$

Now,

$$18000p + 48000p = 66000$$

$$66000p = 66000$$

$$p = 1$$

I). The price of each veg roll = $100p = (100 \times 1) = \text{Rs. } 100$

And, the price of each non-veg roll = $150p = (150 \times 1) = \text{Rs. } 150$

So, the required difference = $(150 - 100) = \text{Rs. } 50$.

Therefore, statement I can be determined.

II). The total revenue generated from the sale of all veg rolls = $18000p = (18000 \times 1) = \text{Rs. } 18000$

Therefore, statement II can be determined.

III). The total revenue generated from the sale of all non-veg rolls = $48000p = (48000 \times 1) = \text{Rs. } 48000$

So, the required percentage = $[\{(48000 - 18000) / 48000\} \times 100]\% = \{(30000 / 48000) \times 100\}\% = 62.50\% \text{ less.}$

Therefore, statement III can be determined.

So, all I, II and III can be determined.

Hence, the correct answer is option D.

2. Questions

Shop A

The total number of rolls made = 600

The total number of veg rolls made = 40% of 600 = 240

The total number of non-veg rolls made = 60% of 600 = 360

Let the number of veg and non-veg rolls sold be $9n$ and $16n$ respectively.

So, the number of veg rolls unsold = $(240 - 9n)$

And, the number of non-veg rolls unsold = $(360 - 16n)$

Now,

$$(240 - 9n) - (360 - 16n) = 20$$

$$220 - 9n = 360 - 16n$$

$$7n = 140$$

$$n = 20$$

Therefore,

The number of veg rolls sold = $9n = (9 \times 20) = 180$

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The number of non-veg rolls sold = $16n = (16 \times 20) = 320$

The number of veg rolls unsold = $(240 - 9n) = (240 - 180) = 60$

The number of non-veg rolls unsold = $(360 - 16n) = (360 - 320) = 40$

Shop B

The total number of non-veg rolls made = $5/9$ of $360 = 200$ (**Note: $44.44\% = 4/9$**)

The total number of veg rolls made = $\{(200/40) \times 60\} = 300$

The total number of rolls made = $(300 + 200) = 500$

Let the number of veg and non-veg rolls sold be $5m$ and $3m$ respectively.

So, the number of veg rolls unsold = $(300 - 5m)$

And, the number of non-veg rolls unsold = $(200 - 3m)$

Now,

$$(300 - 5m) = (200 - 3m)$$

$$2m = 100$$

$$m = 50$$

Therefore,

The number of veg rolls sold = $5m = (5 \times 50) = 250$

The number of non-veg rolls sold = $3m = (3 \times 50) = 150$

The number of veg rolls unsold = $(300 - 5m) = (300 - 250) = 50$

The number of non-veg rolls unsold = $(200 - 3m) = (200 - 150) = 50$

Shop C

The number of veg rolls made = $\{(300/6) \times 7\} = 350$

The number of non-veg rolls made = $\{(350/50) \times 50\} = 350$

The number of total number of rolls made = $(350 + 350) = 700$

Let the number of veg and non-veg rolls sold be $4k$ and $3k$ respectively.

So, the number of non-veg rolls unsold = $(4k/2) = 2k$

And, the number of veg rolls unsold = $(350 - 4k)$

Now,

$$3k + 2k = 350$$

$$5k = 350$$

$$k = 70$$

Therefore,

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The number of veg rolls sold = $4k = (4 \times 70) = 280$

The number of non-veg rolls sold = $3k = (3 \times 70) = 210$

The number of veg rolls unsold = $(350 - 4k) = (350 - 280) = 70$

The number of non-veg rolls unsold = $2k = (2 \times 70) = 140$

Shop D

The total number of rolls made = $(700 + 100) = 800$

The total number of veg rolls made = 80% of 800 = 640

The total number of non-veg rolls made = 20% of 800 = 160

Let the number of veg and non-veg rolls sold be $6j$ and j respectively.

So, the number of veg rolls unsold = $(640 - 6j)$

And, the number of non-veg rolls unsold = $(160 - j)$

Now,

$$(640 - 6j)/j = 2/5$$

$$3200 - 30j = 2j$$

$$32j = 3200$$

$$j = 100$$

Therefore,

The number of veg rolls sold = $6j = (6 \times 100) = 600$

The number of non-veg rolls sold = $j = 100$

The number of veg rolls unsold = $(640 - 6j) = (640 - 600) = 40$

The number of non-veg rolls unsold = $(160 - j) = (160 - 100) = 60$

Veg Rolls

Shop	The total number of veg rolls made	The number of veg rolls sold	The number of veg rolls unsold
A	240	180	60
B	300	250	50
C	350	280	70
D	640	600	40
Total			

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Non-veg Rolls

Shop	The total number of non-veg rolls made	The number of non-veg rolls sold	The number of non-veg rolls unsold
A	360	320	40
B	200	150	50
C	350	210	140
D	160	100	60
Total			

Answer: A

Shop B

The total number of veg rolls sold = 250

Let the number of veg rolls sold to females and girls be $4t$ and t respectively.

The number of veg rolls sold to boys = 30

So, the number of veg rolls sold to males = $(30 \times 4) = 120$

Now,

$$120 + 4t + 30 + t = 250$$

$$5t = 250 - 150$$

$$5t = 100$$

$$t = 20$$

Therefore, the total number of veg rolls sold to adults (males + females) = $\{(120 + 4t) = (120 + 4 \times 20) = (120 + 80) = 200\}$.

So, the required percentage = $\{(200/250) \times 100\} \% = 80\%$.

Hence, the correct answer is option A.

3. Questions

Shop A

The total number of rolls made = 600

The total number of veg rolls made = 40% of 600 = 240

The total number of non-veg rolls made = 60% of 600 = 360

Let the number of veg and non-veg rolls sold be $9n$ and $16n$ respectively.

So, the number of veg rolls unsold = $(240 - 9n)$

And, the number of non-veg rolls unsold = $(360 - 16n)$

Now,

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$$(240-9n) - (360-16n) = 20$$

$$220 - 9n = 360 - 16n$$

$$7n = 140$$

$$n = 20$$

Therefore,

$$\text{The number of veg rolls sold} = 9n = (9 \times 20) = 180$$

$$\text{The number of non-veg rolls sold} = 16n = (16 \times 20) = 320$$

$$\text{The number of veg rolls unsold} = (240-9n) = (240-180) = 60$$

$$\text{The number of non-veg rolls unsold} = (360-16n) = (360-320) = 40$$

Shop B

$$\text{The total number of non-veg rolls made} = 5/9 \text{ of } 360 = 200 \text{ (Note: } 44.44\% = 4/9)$$

$$\text{The total number of veg rolls made} = \{(200/40) \times 60\} = 300$$

$$\text{The total number of rolls made} = (300+200) = 500$$

Let the number of veg and non-veg rolls sold be 5m and 3m respectively.

$$\text{So, the number of veg rolls unsold} = (300-5m)$$

$$\text{And, the number of non-veg rolls unsold} = (200-3m)$$

Now,

$$(300-5m) = (200-3m)$$

$$2m = 100$$

$$m = 50$$

Therefore,

$$\text{The number of veg rolls sold} = 5m = (5 \times 50) = 250$$

$$\text{The number of non-veg rolls sold} = 3m = (3 \times 50) = 150$$

$$\text{The number of veg rolls unsold} = (300-5m) = (300-250) = 50$$

$$\text{The number of non-veg rolls unsold} = (200-3m) = (200-150) = 50$$

Shop C

$$\text{The number of veg rolls made} = \{(300/6) \times 7\} = 350$$

$$\text{The number of non-veg rolls made} = \{(350/50) \times 50\} = 350$$

$$\text{The number of total number of rolls made} = (350+350) = 700$$

Let the number of veg and non-veg rolls sold be 4k and 3k respectively.

$$\text{So, the number of non-veg rolls unsold} = (4k/2) = 2k$$

And, the number of veg rolls unsold = $(350-4k)$

Now,

$$3k + 2k = 350$$

$$5k = 350$$

$$k = 70$$

Therefore,

$$\text{The number of veg rolls sold} = 4k = (4 \times 70) = 280$$

$$\text{The number of non-veg rolls sold} = 3k = (3 \times 70) = 210$$

$$\text{The number of veg rolls unsold} = (350-4k) = (350-280) = 70$$

$$\text{The number of non-veg rolls unsold} = 2k = (2 \times 70) = 140$$

Shop D

$$\text{The total number of rolls made} = (700+100) = 800$$

$$\text{The total number of veg rolls made} = 80\% \text{ of } 800 = 640$$

$$\text{The total number of non-veg rolls made} = 20\% \text{ of } 800 = 160$$

Let the number of veg and non-veg rolls sold be $6j$ and j respectively.

$$\text{So, the number of veg rolls unsold} = (640-6j)$$

$$\text{And, the number of non-veg rolls unsold} = (160-j)$$

Now,

$$(640-6j)/j = 2/5$$

$$3200 - 30j = 2j$$

$$32j = 3200$$

$$j = 100$$

Therefore,

$$\text{The number of veg rolls sold} = 6j = (6 \times 100) = 600$$

$$\text{The number of non-veg rolls sold} = j = 100$$

$$\text{The number of veg rolls unsold} = (640-6j) = (640-600) = 40$$

$$\text{The number of non-veg rolls unsold} = (160-j) = (160-100) = 60$$

Veg Rolls

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Shop	The total number of veg rolls made	The number of veg rolls sold	The number of veg rolls unsold
A	240	180	60
B	300	250	50
C	350	280	70
D	640	600	40
Total			

Non-veg Rolls

Shop	The total number of non-veg rolls made	The number of non-veg rolls sold	The number of non-veg rolls unsold
A	360	320	40
B	200	150	50
C	350	210	140
D	160	100	60
Total			

Answer: B

The total number of rolls (veg + non-veg) sold = $600 + 100 = 700$

Shop D

The total number of pizzas sold = $\frac{6}{7}$ of $700 = 600$ (**Note: $14.28\% = \frac{1}{7}$**)

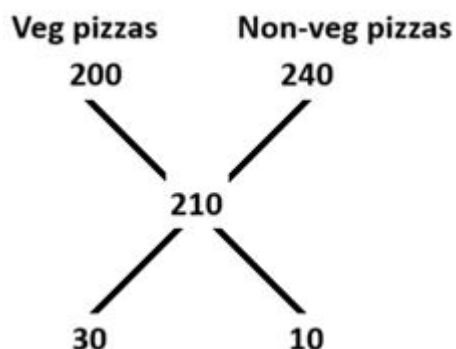
The price of each pizza sold (veg + non-veg) = $(126000/600) = \text{Rs. } 210$

The price of each non-veg pizza = Rs. 240

The price of each veg pizza = $\{(240/120) \times 100\} = \text{Rs. } 200$

Using allegation method, we get,

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So, the ratio of the number of veg and non-veg pizzas sold = $30:10 = 3:1$.

Therefore, the number of non-veg pizzas sold by shop D = $1/4$ of $600 = 150$.

Hence, the correct answer is option B.

4. Questions

Shop A

The total number of rolls made = 600

The total number of veg rolls made = 40% of $600 = 240$

The total number of non-veg rolls made = 60% of $600 = 360$

Let the number of veg and non-veg rolls sold be $9n$ and $16n$ respectively.

So, the number of veg rolls unsold = $(240-9n)$

And, the number of non-veg rolls unsold = $(360-16n)$

Now,

$$(240-9n) - (360-16n) = 20$$

$$220 - 9n = 360 - 16n$$

$$7n = 140$$

$$n = 20$$

Therefore,

The number of veg rolls sold = $9n = (9 \times 20) = 180$

The number of non-veg rolls sold = $16n = (16 \times 20) = 320$

The number of veg rolls unsold = $(240-9n) = (240-180) = 60$

The number of non-veg rolls unsold = $(360-16n) = (360-320) = 40$

Shop B

The total number of non-veg rolls made = $5/9$ of $360 = 200$ (Note: $44.44\% = 4/9$)

The total number of veg rolls made = $\{(200/40) \times 60\} = 300$

The total number of rolls made = $(300+200) = 500$

Let the number of veg and non-veg rolls sold be $5m$ and $3m$ respectively.

So, the number of veg rolls unsold = $(300-5m)$

And, the number of non-veg rolls unsold = $(200-3m)$

Now,

$$(300-5m) = (200-3m)$$

$$2m = 100$$

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$$m = 50$$

Therefore,

$$\text{The number of veg rolls sold} = 5m = (5 \times 50) = 250$$

$$\text{The number of non-veg rolls sold} = 3m = (3 \times 50) = 150$$

$$\text{The number of veg rolls unsold} = (300 - 5m) = (300 - 250) = 50$$

$$\text{The number of non-veg rolls unsold} = (200 - 3m) = (200 - 150) = 50$$

Shop C

$$\text{The number of veg rolls made} = \{(300/6) \times 7\} = 350$$

$$\text{The number of non-veg rolls made} = \{(350/50) \times 50\} = 350$$

$$\text{The number of total number of rolls made} = (350 + 350) = 700$$

Let the number of veg and non-veg rolls sold be $4k$ and $3k$ respectively.

$$\text{So, the number of non-veg rolls unsold} = (4k/2) = 2k$$

$$\text{And, the number of veg rolls unsold} = (350 - 4k)$$

Now,

$$3k + 2k = 350$$

$$5k = 350$$

$$k = 70$$

Therefore,

$$\text{The number of veg rolls sold} = 4k = (4 \times 70) = 280$$

$$\text{The number of non-veg rolls sold} = 3k = (3 \times 70) = 210$$

$$\text{The number of veg rolls unsold} = (350 - 4k) = (350 - 280) = 70$$

$$\text{The number of non-veg rolls unsold} = 2k = (2 \times 70) = 140$$

Shop D

$$\text{The total number of rolls made} = (700 + 100) = 800$$

$$\text{The total number of veg rolls made} = 80\% \text{ of } 800 = 640$$

$$\text{The total number of non-veg rolls made} = 20\% \text{ of } 800 = 160$$

Let the number of veg and non-veg rolls sold be $6j$ and j respectively.

$$\text{So, the number of veg rolls unsold} = (640 - 6j)$$

$$\text{And, the number of non-veg rolls unsold} = (160 - j)$$

Now,

$$(640 - 6j)/j = 2/5$$

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$$3200 - 30j = 2j$$

$$32j = 3200$$

$$j = 100$$

Therefore,

$$\text{The number of veg rolls sold} = 6j = (6 \times 100) = 600$$

$$\text{The number of non-veg rolls sold} = j = 100$$

$$\text{The number of veg rolls unsold} = (640 - 6j) = (640 - 600) = 40$$

$$\text{The number of non-veg rolls unsold} = (160 - j) = (160 - 100) = 60$$

Veg Rolls

Shop	The total number of veg rolls made	The number of veg rolls sold	The number of veg rolls unsold
A	240	180	60
B	300	250	50
C	350	280	70
D	640	600	40
Total			

Non-veg Rolls

Shop	The total number of non-veg rolls made	The number of non-veg rolls sold	The number of non-veg rolls unsold
A	360	320	40
B	200	150	50
C	350	210	140
D	160	100	60
Total			

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Answer: D

Shop C

$$\text{The total number of veg rolls sold} = 280$$

Let the number of tomato and onion veg rolls sold be 9s and 7s respectively.

$$\text{And, the number of cheese veg rolls sold} = (9s + 30)$$

Now,

$$(9s+30) + 9s + 7s = 280$$

$$25s = 250$$

$$s = 10$$

Therefore, the number of onion veg rolls sold by shop C = $7s = (7 \times 10) = 70$.

Hence, the correct answer is option D.

5. Questions

Shop A

The total number of rolls made = 600

The total number of veg rolls made = 40% of 600 = 240

The total number of non-veg rolls made = 60% of 600 = 360

Let the number of veg and non-veg rolls sold be $9n$ and $16n$ respectively.

So, the number of veg rolls unsold = $(240 - 9n)$

And, the number of non-veg rolls unsold = $(360 - 16n)$

Now,

$$(240 - 9n) - (360 - 16n) = 20$$

$$220 - 9n = 360 - 16n$$

$$7n = 140$$

$$n = 20$$

Therefore,

The number of veg rolls sold = $9n = (9 \times 20) = 180$

The number of non-veg rolls sold = $16n = (16 \times 20) = 320$

The number of veg rolls unsold = $(240 - 9n) = (240 - 180) = 60$

The number of non-veg rolls unsold = $(360 - 16n) = (360 - 320) = 40$

Shop B

The total number of non-veg rolls made = $5/9$ of 360 = 200 (Note: $44.44\% = 4/9$)

The total number of veg rolls made = $\{(200/40) \times 60\} = 300$

The total number of rolls made = $(300 + 200) = 500$

Let the number of veg and non-veg rolls sold be $5m$ and $3m$ respectively.

So, the number of veg rolls unsold = $(300 - 5m)$

And, the number of non-veg rolls unsold = $(200 - 3m)$

Now,

$$(300-5m) = (200-3m)$$

$$2m = 100$$

$$m = 50$$

Therefore,

$$\text{The number of veg rolls sold} = 5m = (5 \times 50) = 250$$

$$\text{The number of non-veg rolls sold} = 3m = (3 \times 50) = 150$$

$$\text{The number of veg rolls unsold} = (300-5m) = (300-250) = 50$$

$$\text{The number of non-veg rolls unsold} = (200-3m) = (200-150) = 50$$

Shop C

$$\text{The number of veg rolls made} = \{(300/6) \times 7\} = 350$$

$$\text{The number of non-veg rolls made} = \{(350/50) \times 50\} = 350$$

$$\text{The number of total number of rolls made} = (350+350) = 700$$

Let the number of veg and non-veg rolls sold be $4k$ and $3k$ respectively.

$$\text{So, the number of non-veg rolls unsold} = (4k/2) = 2k$$

$$\text{And, the number of veg rolls unsold} = (350-4k)$$

Now,

$$3k + 2k = 350$$

$$5k = 350$$

$$k = 70$$

Therefore,

$$\text{The number of veg rolls sold} = 4k = (4 \times 70) = 280$$

$$\text{The number of non-veg rolls sold} = 3k = (3 \times 70) = 210$$

$$\text{The number of veg rolls unsold} = (350-4k) = (350-280) = 70$$

$$\text{The number of non-veg rolls unsold} = 2k = (2 \times 70) = 140$$

Shop D

$$\text{The total number of rolls made} = (700+100) = 800$$

$$\text{The total number of veg rolls made} = 80\% \text{ of } 800 = 640$$

$$\text{The total number of non-veg rolls made} = 20\% \text{ of } 800 = 160$$

Let the number of veg and non-veg rolls sold be $6j$ and j respectively.

$$\text{So, the number of veg rolls unsold} = (640-6j)$$

$$\text{And, the number of non-veg rolls unsold} = (160-j)$$

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Now,

$$(640-6j)/j = 2/5$$

$$3200 - 30j = 2j$$

$$32j = 3200$$

$$j = 100$$

Therefore,

$$\text{The number of veg rolls sold} = 6j = (6 \times 100) = 600$$

$$\text{The number of non-veg rolls sold} = j = 100$$

$$\text{The number of veg rolls unsold} = (640-6j) = (640-600) = 40$$

$$\text{The number of non-veg rolls unsold} = (160-j) = (160-100) = 60$$

Veg Rolls

Shop	The total number of veg rolls made	The number of veg rolls sold	The number of veg rolls unsold
A	240	180	60
B	300	250	50
C	350	280	70
D	640	600	40
Total			

Non-veg Rolls

Shop	The total number of non-veg rolls made	The number of non-veg rolls sold	The number of non-veg rolls unsold
A	360	320	40
B	200	150	50
C	350	210	140
D	160	100	60
Total			

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Answer: A

The total number of rolls (veg + non-veg) made by shop B = 500

Shop E

The total number of rolls (veg + non-veg) made = 500

The number of non-veg rolls unsold = 10

The number of veg rolls unsold = $(10 \times 10) = 100$

Let the number of veg rolls sold be L.

So, the number of non-veg rolls sold = $(L - 210)$

Now,

$$L + (L - 210) + 10 + 100 = 500$$

$$2L - 210 = 390$$

$$2L = 600$$

$$L = 300$$

Therefore, the number of veg rolls sold = $L = 300$.

Hence, the correct answer is option A.

6. Questions

The ratio of the number of Indian males and females who visited each place,

$$\text{Taj Mahal} = 1.50 = 1.50:1 = 3:2$$

$$\text{Red Fort} = 1.67 = 1.67:1 = 5:3$$

$$\text{India Gate} = 1.33 = 1.33:1 = 4:3$$

$$\text{Hampi} = 2 = 2:1$$

The ratio of the number of foreigners males and females who visited each place,

$$\text{Taj Mahal} = 1 = 1:1$$

$$\text{Red Fort} = 2 = 2:1$$

$$\text{India Gate} = 1.50 = 1.50:1 = 3:2$$

$$\text{Hampi} = 1.25 = 1.25:1 = 5:4$$

Taj Mahal

The total number of visitors who visited = 700

The total number of males (Indian + Foreigners) who visited = $\frac{4}{7}$ of 700 = 400

The total number of females (Indian + Foreigners) who visited = $\frac{3}{7}$ of 700 = 300

Now, let the number of Indian males and females who visited be $3a$ and $2a$ respectively.

And, let the number of foreigner males and females who visited be b and b respectively.

So,

$$3a + b = 400 \dots\dots\dots(I)$$

$$2a + b = 300 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$6a + 2b = 800$$

$$6a + 3b = 900$$

$$b = 100$$

Now, put the value of b in equation (I), we get,

$$3a + b = 400$$

$$3a + 100 = 400$$

$$3a = 300$$

$$a = 100$$

Red Fort

The total number of males (Indian + Foreigners) who visited = 450

The total number of females (Indian + Foreigners) who visited = $\{(450/180)*100\} = 250$

The total number of visitors who visited = $(450+250) = 700$

Now, let the number of Indian males and females who visited be 5c and 3c respectively.

And, let the number of foreigner males and females who visited be 2d and d respectively.

So,

$$5c + 2d = 450 \dots \dots \dots (III)$$

$$3c + d = 250 \dots \dots \dots (IV)$$

By solving equation (III) and (IV), we get,

$$15c + 6d = 1350$$

$$15c + 5d = 1250$$

$$d = 100$$

By solving equation (III) and (IV), we get,

$$5c + 2d = 450$$

$$5c + (2*100) = 450$$

$$5c = 450 - 200$$

$$5c = 250$$

$$c = 50$$

India Gate

The total number of visitors who visited = 500

The total number of females (Indian + Foreigners) who visited = $(400-190) = 210$

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The total number of males (Indian + Foreigners) who visited = $(500-210) = 290$

Now, let the number of Indian males and females who visited be $4e$ and $3e$ respectively.

And, let the number of foreigner males and females who visited be $3f$ and $2f$ respectively.

So,

$$4e + 3f = 290 \dots\dots\dots(V)$$

$$3e + 2f = 210 \dots\dots\dots(VI)$$

By solving equation (V) and (VI), we get,

$$12e + 9f = 870$$

$$12e + 8f = 840$$

$$f = 30$$

Now, put the value of f in equation (V), we get,

$$4e + (3 \times 30) = 290$$

$$4e = 290 - 90$$

$$4e = 200$$

$$e = 50$$

Hampi

The total number of visitors who visited = $(500+550) = 1050$

The total number of females (Indian + Foreigners) who visited = 400

The total number of males (Indian + Foreigners) who visited = $(1050-400) = 650$

Now, let the number of Indian males and females who visited be $2g$ and g respectively.

And, let the number of foreigner males and females who visited be $5h$ and $4h$ respectively.

So,

$$2g + 5h = 650 \dots\dots\dots(VII)$$

$$g + 4h = 400 \dots\dots\dots(VIII)$$

By solving equation (VII) and (VIII), we get,

$$2g + 5h = 650$$

$$2g + 8h = 800$$

$$3h = 150$$

$$h = 50$$

By solving equation (VII) and (VIII), we get,

$$2g + 5h = 650$$

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$$2g + (5 \times 50) = 650$$

$$2g = 650 - 250$$

$$2g = 400$$

$$g = 200$$

Taj Mahal

$$\text{The number of Indian males who visited} = 3a = (3 \times 100) = 300$$

$$\text{The number of Indian females who visited} = 2a = (2 \times 100) = 200$$

$$\text{The number of foreigner males who visited} = b = 100$$

$$\text{The number of foreigner females who visited} = b = 100$$

$$\text{So, the total number of Indians who visited} = (300 + 200) = 500$$

$$\text{And, the total number of foreigners who visited} = (100 + 100) = 200$$

Red Fort

$$\text{The number of Indian males who visited} = 5c = (5 \times 50) = 250$$

$$\text{The number of Indian females who visited} = 3c = (3 \times 50) = 150$$

$$\text{The number of foreigner males who visited} = 2d = (2 \times 100) = 200$$

$$\text{The number of foreigner females who visited} = d = 100$$

$$\text{So, the total number of Indians who visited} = (250 + 150) = 400$$

$$\text{And, the total number of foreigners who visited} = (200 + 100) = 300$$

India Gate

$$\text{The number of Indian males who visited} = 4e = (4 \times 50) = 200$$

$$\text{The number of Indian females who visited} = 3e = (3 \times 50) = 150$$

$$\text{The number of foreigner males who visited} = 3f = (3 \times 30) = 90$$

$$\text{The number of foreigner females who visited} = 2f = (2 \times 30) = 60$$

$$\text{So, the total number of Indians who visited} = (200 + 150) = 350$$

$$\text{And, the total number of foreigners who visited} = (90 + 60) = 150$$

Hampi

$$\text{The number of Indian males who visited} = 2g = (2 \times 200) = 400$$

$$\text{The number of Indian females who visited} = g = 200$$

$$\text{The number of foreigner males who visited} = 5h = (5 \times 50) = 250$$

$$\text{The number of foreigner females who visited} = 4h = (4 \times 50) = 200$$

$$\text{So, the total number of Indians who visited} = (400 + 200) = 600$$

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And, the total number of foreigners who visited = $(250+200) = 450$

Indians

Name of the place	The total number of Indian visitors who visited	The number of Indian males who visited	The number of Indian females who visited
Taj Mahal	500	300	200
Red Fort	400	250	150
India Gate	350	200	150
Hampi	600	400	200
Total	1850	1150	700

Foreigners

Name of the place	The total number of foreigner visitors who visited	The number of foreigner males who visited	The number of foreigner females who visited
Taj Mahal	200	100	100
Red Fort	300	200	100
India Gate	150	90	60
Hampi	450	250	200
Total	1100	640	460

Answer: A

Now, according to the question,

Taj Mahal

The total number of Indians who visited = 500

Let the total number of employed Indians who visited be 100p

So, the number of unemployed Indians who visited = $(500-100p)$

The number of graduate Indians who visited = 80% of 100p = 80p

The number of post-graduate Indians who visited = 20% of 100p = 20p

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Now,

$$(500-100p) - 20p = 20$$

$$500 - 120p = 20$$

$$120p = 480$$

$$p = 4$$

So, the number of employed Indians who visited = $100p = (100 \times 4) = 400$.

And, the number of unemployed Indians who visited = $(500-100p) = (500-400) = 100$.

Therefore, the required ratio = $400:100 = 4:1$.

Hence, the correct answer is option A.

7. Questions

The ratio of the number of Indian males and females who visited each place,

$$\text{Taj Mahal} = 1.50 = 1.50:1 = 3:2$$

$$\text{Red Fort} = 1.67 = 1.67:1 = 5:3$$

$$\text{India Gate} = 1.33 = 1.33:1 = 4:3$$

$$\text{Hampi} = 2 = 2:1$$

The ratio of the number of foreigners males and females who visited each place,

$$\text{Taj Mahal} = 1 = 1:1$$

$$\text{Red Fort} = 2 = 2:1$$

$$\text{India Gate} = 1.50 = 1.50:1 = 3:2$$

$$\text{Hampi} = 1.25 = 1.25:1 = 5:4$$

Taj Mahal

The total number of visitors who visited = 700

The total number of males(Indian + Foreigners) who visited = $\frac{4}{7}$ of 700 = 400

The total number of females(Indian + Foreigners) who visited = $\frac{3}{7}$ of 700 = 300

Now, let the number of Indian males and females who visited be $3a$ and $2a$ respectively.

And, let the number of foreigner males and females who visited be b and b respectively.

So,

$$3a + b = 400 \dots\dots\dots(I)$$

$$2a + b = 300 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$6a + 2b = 800$$

$$6a + 3b = 900$$

$$b = 100$$

Now, put the value of b in equation (I), we get,

$$3a + b = 400$$

$$3a + 100 = 400$$

$$3a = 300$$

$$a = 100$$

Red Fort

The total number of males (Indian + Foreigners) who visited = 450

The total number of females (Indian + Foreigners) who visited = $\{(450/180)*100\} = 250$

The total number of visitors who visited = $(450+250) = 700$

Now, let the number of Indian males and females who visited be 5c and 3c respectively.

And, let the number of foreigner males and females who visited be 2d and d respectively.

So,

$$5c + 2d = 450 \dots \dots \dots (III)$$

$$3c + d = 250 \dots \dots \dots (IV)$$

By solving equation (III) and (IV), we get,

$$15c + 6d = 1350$$

$$15c + 5d = 1250$$

$$d = 100$$

By solving equation (III) and (IV), we get,

$$5c + 2d = 450$$

$$5c + (2*100) = 450$$

$$5c = 450 - 200$$

$$5c = 250$$

$$c = 50$$

India Gate

The total number of visitors who visited = 500

The total number of females (Indian + Foreigners) who visited = $(400-190) = 210$

The total number of males (Indian + Foreigners) who visited = $(500-210) = 290$

Now, let the number of Indian males and females who visited be 4e and 3e respectively.

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And, let the number of foreigner males and females who visited be $3f$ and $2f$ respectively.

So,

$$4e + 3f = 290 \dots\dots\dots(V)$$

$$3e + 2f = 210 \dots\dots\dots(VI)$$

By solving equation (V) and (VI), we get,

$$12e + 9f = 870$$

$$12e + 8f = 840$$

$$f = 30$$

Now, put the value of f in equation (V), we get,

$$4e + (3 \times 30) = 290$$

$$4e = 290 - 90$$

$$4e = 200$$

$$e = 50$$

Hampi

The total number of visitors who visited = $(500+550) = 1050$

The total number of females (Indian + Foreigners) who visited = 400

The total number of males (Indian + Foreigners) who visited = $(1050-400) = 650$

Now, let the number of Indian males and females who visited be $2g$ and g respectively.

And, let the number of foreigner males and females who visited be $5h$ and $4h$ respectively.

So,

$$2g + 5h = 650 \dots\dots\dots(VII)$$

$$g + 4h = 400 \dots\dots\dots(VIII)$$

By solving equation (VII) and (VIII), we get,

$$2g + 5h = 650$$

$$2g + 8h = 800$$

$$3h = 150$$

$$h = 50$$

By solving equation (VII) and (VIII), we get,

$$2g + 5h = 650$$

$$2g + (5 \times 50) = 650$$

$$2g = 650 - 250$$

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$$2g = 400$$

$$g = 200$$

Taj Mahal

$$\text{The number of Indian males who visited} = 3a = (3 \times 100) = 300$$

$$\text{The number of Indian females who visited} = 2a = (2 \times 100) = 200$$

$$\text{The number of foreigner males who visited} = b = 100$$

$$\text{The number of foreigner females who visited} = b = 100$$

$$\text{So, the total number of Indians who visited} = (300 + 200) = 500$$

$$\text{And, the total number of foreigners who visited} = (100 + 100) = 200$$

Red Fort

$$\text{The number of Indian males who visited} = 5c = (5 \times 50) = 250$$

$$\text{The number of Indian females who visited} = 3c = (3 \times 50) = 150$$

$$\text{The number of foreigner males who visited} = 2d = (2 \times 100) = 200$$

$$\text{The number of foreigner females who visited} = d = 100$$

$$\text{So, the total number of Indians who visited} = (250 + 150) = 400$$

$$\text{And, the total number of foreigners who visited} = (200 + 100) = 300$$

India Gate

$$\text{The number of Indian males who visited} = 4e = (4 \times 50) = 200$$

$$\text{The number of Indian females who visited} = 3e = (3 \times 50) = 150$$

$$\text{The number of foreigner males who visited} = 3f = (3 \times 30) = 90$$

$$\text{The number of foreigner females who visited} = 2f = (2 \times 30) = 60$$

$$\text{So, the total number of Indians who visited} = (200 + 150) = 350$$

$$\text{And, the total number of foreigners who visited} = (90 + 60) = 150$$

Hampi

$$\text{The number of Indian males who visited} = 2g = (2 \times 200) = 400$$

$$\text{The number of Indian females who visited} = g = 200$$

$$\text{The number of foreigner males who visited} = 5h = (5 \times 50) = 250$$

$$\text{The number of foreigner females who visited} = 4h = (4 \times 50) = 200$$

$$\text{So, the total number of Indians who visited} = (400 + 200) = 600$$

$$\text{And, the total number of foreigners who visited} = (250 + 200) = 450$$

Indians

@QuizByMickey

Name of the place	The total number of Indian visitors who visited	The number of Indian males who visited	The number of Indian females who visited
Taj Mahal	500	300	200
Red Fort	400	250	150
India Gate	350	200	150
Hampi	600	400	200
Total	1850	1150	700

Foreigners

Name of the place	The total number of foreigner visitors who visited	The number of foreigner males who visited	The number of foreigner females who visited
Taj Mahal	200	100	100
Red Fort	300	200	100
India Gate	150	90	60
Hampi	450	250	200
Total	1100	640	460

Answer: C

Red Fort

The total number of foreigner males who visited (UK + USA) = 200

The total number of foreigner females who visited (UK + USA) = 100

Using I

The number of foreigner females from UK who visited = 90

The number of foreigner males from UK who visited = $\{(90/50)*100\} = 180$

So, the number of foreigner females from USA who visited = $(100-90) = 10$

And, the number of foreigner males from USA who visited = $(200-180) = 20$

Therefore, the total number of foreigners (males + females) from USA who visited = $(20+10) = 30$.

So, I does satisfy.

Using II

The number of foreigner females from UK who visited = 80

The number of foreigner males from UK who visited = $\{(80/50)*100\} = 160$

So, the number of foreigner females from USA who visited = $(100-80) = 20$

And, the number of foreigner males from USA who visited = $(200-160) = 40$

Therefore, the total number of foreigners (males + females) from USA who visited = $(40+20) = 60$.

So, II does not satisfy.

Using III

The number of foreigner females from UK who visited = 70

The number of foreigner males from UK who visited = $\{(70/50)*100\} = 140$

So, the number of foreigner females from USA who visited = $(100-70) = 30$

And, the number of foreigner males from USA who visited = $(200-140) = 60$

Therefore, the total number of foreigners (males + females) from USA who visited = $(60+30) = 90$.

So, III does not satisfy.

Therefore, only I follow.

Hence, the correct answer is option C.

8. Questions

The ratio of the number of Indian males and females who visited each place,

Taj Mahal = 1.50 = 1.50:1 = 3:2

Red Fort = 1.67 = 1.67:1 = 5:3

India Gate = 1.33 = 1.33:1 = 4:3

Hampi = 2 = 2:1

The ratio of the number of foreigners males and females who visited each place,

Taj Mahal = 1 = 1:1

Red Fort = 2 = 2:1

India Gate = 1.50 = 1.50:1 = 3:2

Hampi = 1.25 = 1.25:1 = 5:4

Taj Mahal

The total number of visitors who visited = 700

The total number of males(Indian + Foreigners) who visited = $4/7$ of 700 = 400

The total number of females(Indian + Foreigners) who visited = $\frac{3}{7}$ of 700 = 300

Now, let the number of Indian males and females who visited be $3a$ and $2a$ respectively.

And, let the number of foreigner males and females who visited be b and b respectively.

So,

$$3a + b = 400 \dots\dots\dots(I)$$

$$2a + b = 300 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$6a + 2b = 800$$

$$6a + 3b = 900$$

$$b = 100$$

Now, put the value of b in equation (I), we get,

$$3a + b = 400$$

$$3a + 100 = 400$$

$$3a = 300$$

$$a = 100$$

Red Fort

The total number of males (Indian + Foreigners) who visited = 450

The total number of females (Indian + Foreigners) who visited = $\{(450/180)*100\} = 250$

The total number of visitors who visited = $(450+250) = 700$

Now, let the number of Indian males and females who visited be $5c$ and $3c$ respectively.

And, let the number of foreigner males and females who visited be $2d$ and d respectively.

So,

$$5c + 2d = 450 \dots\dots\dots(III)$$

$$3c + d = 250 \dots\dots\dots(IV)$$

By solving equation (III) and (IV), we get,

$$15c + 6d = 1350$$

$$15c + 5d = 1250$$

$$d = 100$$

By solving equation (III) and (IV), we get,

$$5c + 2d = 450$$

$$5c + (2*100) = 450$$

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$$5c = 450 - 200$$

$$5c = 250$$

$$c = 50$$

India Gate

The total number of visitors who visited = 500

The total number of females (Indian + Foreigners) who visited = $(400-190) = 210$

The total number of males (Indian + Foreigners) who visited = $(500-210) = 290$

Now, let the number of Indian males and females who visited be $4e$ and $3e$ respectively.

And, let the number of foreigner males and females who visited be $3f$ and $2f$ respectively.

So,

$$4e + 3f = 290 \dots\dots\dots(V)$$

$$3e + 2f = 210 \dots\dots\dots(VI)$$

By solving equation (V) and (VI), we get,

$$12e + 9f = 870$$

$$12e + 8f = 840$$

$$f = 30$$

Now, put the value of f in equation (V), we get,

$$4e + (3 \times 30) = 290$$

$$4e = 290 - 90$$

$$4e = 200$$

$$e = 50$$

Hampi

The total number of visitors who visited = $(500+550) = 1050$

The total number of females (Indian + Foreigners) who visited = 400

The total number of males (Indian + Foreigners) who visited = $(1050-400) = 650$

Now, let the number of Indian males and females who visited be $2g$ and g respectively.

And, let the number of foreigner males and females who visited be $5h$ and $4h$ respectively.

So,

$$2g + 5h = 650 \dots\dots\dots(VII)$$

$$g + 4h = 400 \dots\dots\dots(VIII)$$

By solving equation (VII) and (VIII), we get,

$$2g + 5h = 650$$

$$2g + 8h = 800$$

$$3h = 150$$

$$h = 50$$

By solving equation (VII) and (VIII), we get,

$$2g + 5h = 650$$

$$2g + (5 \times 50) = 650$$

$$2g = 650 - 250$$

$$2g = 400$$

$$g = 200$$

Taj Mahal

The number of Indian males who visited = $3a = (3 \times 100) = 300$

The number of Indian females who visited = $2a = (2 \times 100) = 200$

The number of foreigner males who visited = $b = 100$

The number of foreigner females who visited = $b = 100$

So, the total number of Indians who visited = $(300 + 200) = 500$

And, the total number of foreigners who visited = $(100 + 100) = 200$

Red Fort

The number of Indian males who visited = $5c = (5 \times 50) = 250$

The number of Indian females who visited = $3c = (3 \times 50) = 150$

The number of foreigner males who visited = $2d = (2 \times 100) = 200$

The number of foreigner females who visited = $d = 100$

So, the total number of Indians who visited = $(250 + 150) = 400$

And, the total number of foreigners who visited = $(200 + 100) = 300$

India Gate

The number of Indian males who visited = $4e = (4 \times 50) = 200$

The number of Indian females who visited = $3e = (3 \times 50) = 150$

The number of foreigner males who visited = $3f = (3 \times 30) = 90$

The number of foreigner females who visited = $2f = (2 \times 30) = 60$

So, the total number of Indians who visited = $(200 + 150) = 350$

And, the total number of foreigners who visited = $(90 + 60) = 150$

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Hampi

The number of Indian males who visited = $2g = (2 \times 200) = 400$

The number of Indian females who visited = $g = 200$

The number of foreigner males who visited = $5h = (5 \times 50) = 250$

The number of foreigner females who visited = $4h = (4 \times 50) = 200$

So, the total number of Indians who visited = $(400+200) = 600$

And, the total number of foreigners who visited = $(250+200) = 450$

Indians

Name of the place	The total number of Indian visitors who visited	The number of Indian males who visited	The number of Indian females who visited
Taj Mahal	500	300	200
Red Fort	400	250	150
India Gate	350	200	150
Hampi	600	400	200
Total	1850	1150	700

Foreigners

Name of the place	The total number of foreigner visitors who visited	The number of foreigner males who visited	The number of foreigner females who visited
Taj Mahal	200	100	100
Red Fort	300	200	100
India Gate	150	90	60
Hampi	450	250	200
Total	1100	640	460

Answer: B

India Gate

The total number of Indian males who visited = 200

The total number of Indian females who visited = 150

Let the number of Indian males and females who are illiterate who visited be Q each.

So, the number of Indian males who are literate who visited = $(200-Q)$

And, the number of Indian females who are literate who visited = $(150-Q)$

Now,

$$(200-Q)/(150-Q) = 140/100$$

$$(200-Q)/(150-Q) = 7/5$$

$$1000 - 5Q = 1050 - 7Q$$

$$2Q = 50$$

$$Q = 25$$

Therefore, the total number of literate Indians (males + females) who visited india gate = $\{(200-Q)+(150-Q)\} = \{(200-25)+(150-25)\} = (175+125) = 300$.

Hence, the correct answer is option B.

9. Questions

The ratio of the number of Indian males and females who visited each place,

$$\text{Taj Mahal} = 1.50 = 1.50:1 = 3:2$$

$$\text{Red Fort} = 1.67 = 1.67:1 = 5:3$$

$$\text{India Gate} = 1.33 = 1.33:1 = 4:3$$

$$\text{Hampi} = 2 = 2:1$$

The ratio of the number of foreigners males and females who visited each place,

$$\text{Taj Mahal} = 1 = 1:1$$

$$\text{Red Fort} = 2 = 2:1$$

$$\text{India Gate} = 1.50 = 1.50:1 = 3:2$$

$$\text{Hampi} = 1.25 = 1.25:1 = 5:4$$

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Taj Mahal

The total number of visitors who visited = 700

The total number of males(Indian + Foreigners) who visited = $4/7$ of 700 = 400

The total number of females(Indian + Foreigners) who visited = $3/7$ of 700 = 300

Now, let the number of Indian males and females who visited be 3a and 2a respectively.

And, let the number of foreigner males and females who visited be b and b respectively.

So,

$$3a + b = 400 \dots\dots\dots(I)$$

$$2a + b = 300 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$6a + 2b = 800$$

$$6a + 3b = 900$$

$$b = 100$$

Now, put the value of b in equation (I), we get,

$$3a + b = 400$$

$$3a + 100 = 400$$

$$3a = 300$$

$$a = 100$$

Red Fort

The total number of males (Indian + Foreigners) who visited = 450

The total number of females (Indian + Foreigners) who visited = $\{(450/180)*100\} = 250$

The total number of visitors who visited = $(450+250) = 700$

Now, let the number of Indian males and females who visited be 5c and 3c respectively.

And, let the number of foreigner males and females who visited be 2d and d respectively.

So,

$$5c + 2d = 450 \dots\dots\dots(III)$$

$$3c + d = 250 \dots\dots\dots(IV)$$

By solving equation (III) and (IV), we get,

$$15c + 6d = 1350$$

$$15c + 5d = 1250$$

$$d = 100$$

By solving equation (III) and (IV), we get,

$$5c + 2d = 450$$

$$5c + (2*100) = 450$$

$$5c = 450 - 200$$

$$5c = 250$$

$$c = 50$$

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India Gate

The total number of visitors who visited = 500

The total number of females (Indian + Foreigners) who visited = $(400-190) = 210$

The total number of males (Indian + Foreigners) who visited = $(500-210) = 290$

Now, let the number of Indian males and females who visited be $4e$ and $3e$ respectively.

And, let the number of foreigner males and females who visited be $3f$ and $2f$ respectively.

So,

$$4e + 3f = 290 \dots\dots\dots(V)$$

$$3e + 2f = 210 \dots\dots\dots(VI)$$

By solving equation (V) and (VI), we get,

$$12e + 9f = 870$$

$$12e + 8f = 840$$

$$f = 30$$

Now, put the value of f in equation (V), we get,

$$4e + (3 \times 30) = 290$$

$$4e = 290 - 90$$

$$4e = 200$$

$$e = 50$$

Hampi

The total number of visitors who visited = $(500+550) = 1050$

The total number of females (Indian + Foreigners) who visited = 400

The total number of males (Indian + Foreigners) who visited = $(1050-400) = 650$

Now, let the number of Indian males and females who visited be $2g$ and g respectively.

And, let the number of foreigner males and females who visited be $5h$ and $4h$ respectively.

So,

$$2g + 5h = 650 \dots\dots\dots(VII)$$

$$g + 4h = 400 \dots\dots\dots(VIII)$$

By solving equation (VII) and (VIII), we get,

$$2g + 5h = 650$$

$$2g + 8h = 800$$

$$3h = 150$$

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$$h = 50$$

By solving equation (VII) and (VIII), we get,

$$2g + 5h = 650$$

$$2g + (5 \times 50) = 650$$

$$2g = 650 - 250$$

$$2g = 400$$

$$g = 200$$

Taj Mahal

$$\text{The number of Indian males who visited} = 3a = (3 \times 100) = 300$$

$$\text{The number of Indian females who visited} = 2a = (2 \times 100) = 200$$

$$\text{The number of foreigner males who visited} = b = 100$$

$$\text{The number of foreigner females who visited} = b = 100$$

$$\text{So, the total number of Indians who visited} = (300 + 200) = 500$$

$$\text{And, the total number of foreigners who visited} = (100 + 100) = 200$$

Red Fort

$$\text{The number of Indian males who visited} = 5c = (5 \times 50) = 250$$

$$\text{The number of Indian females who visited} = 3c = (3 \times 50) = 150$$

$$\text{The number of foreigner males who visited} = 2d = (2 \times 100) = 200$$

$$\text{The number of foreigner females who visited} = d = 100$$

$$\text{So, the total number of Indians who visited} = (250 + 150) = 400$$

$$\text{And, the total number of foreigners who visited} = (200 + 100) = 300$$

India Gate

$$\text{The number of Indian males who visited} = 4e = (4 \times 50) = 200$$

$$\text{The number of Indian females who visited} = 3e = (3 \times 50) = 150$$

$$\text{The number of foreigner males who visited} = 3f = (3 \times 30) = 90$$

$$\text{The number of foreigner females who visited} = 2f = (2 \times 30) = 60$$

$$\text{So, the total number of Indians who visited} = (200 + 150) = 350$$

$$\text{And, the total number of foreigners who visited} = (90 + 60) = 150$$

Hampi

$$\text{The number of Indian males who visited} = 2g = (2 \times 200) = 400$$

$$\text{The number of Indian females who visited} = g = 200$$

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The number of foreigner males who visited = $5h = (5 \times 50) = 250$

The number of foreigner females who visited = $4h = (4 \times 50) = 200$

So, the total number of Indians who visited = $(400 + 200) = 600$

And, the total number of foreigners who visited = $(250 + 200) = 450$

Indians

Name of the place	The total number of Indian visitors who visited	The number of Indian males who visited	The number of Indian females who visited
Taj Mahal	500	300	200
Red Fort	400	250	150
India Gate	350	200	150
Hampi	600	400	200
Total	1850	1150	700

Foreigners

Name of the place	The total number of foreigner visitors who visited	The number of foreigner males who visited	The number of foreigner females who visited
Taj Mahal	200	100	100
Red Fort	300	200	100
India Gate	150	90	60
Hampi	450	250	200
Total	1100	640	460

Answer: C

Konark Sun Temple

The total number of foreigners (males + females) who visited = 300

The total number of Indians (males + females) who visited = $\{(300/60) \times 100\} = 500$

Now, let the number of Indian males and foreigner females who visited be $3k$ and k respectively.

So, the number of Indian females who visited = $(500-3k)$

And, the number of foreigners males who visited = $(300-k)$

Now,

$$(500-3k) = (300-k)$$

$$2k = 200$$

$$k = 100$$

So, the number of Indian males who visited = $3k = (3 \times 100) = 300$

Therefore, the required percentage = $\{(300/500) \times 100\} \% = 60\%$.

Hence, the correct answer is option C.

10. Questions

The ratio of the number of Indian males and females who visited each place,

$$\text{Taj Mahal} = 1.50 = 1.50:1 = 3:2$$

$$\text{Red Fort} = 1.67 = 1.67:1 = 5:3$$

$$\text{India Gate} = 1.33 = 1.33:1 = 4:3$$

$$\text{Hampi} = 2 = 2:1$$

The ratio of the number of foreigners males and females who visited each place,

$$\text{Taj Mahal} = 1 = 1:1$$

$$\text{Red Fort} = 2 = 2:1$$

$$\text{India Gate} = 1.50 = 1.50:1 = 3:2$$

$$\text{Hampi} = 1.25 = 1.25:1 = 5:4$$

Taj Mahal

The total number of visitors who visited = 700

The total number of males(Indian + Foreigners) who visited = $4/7$ of 700 = 400

The total number of females(Indian + Foreigners) who visited = $3/7$ of 700 = 300

Now, let the number of Indian males and females who visited be $3a$ and $2a$ respectively.

And, let the number of foreigner males and females who visited be b and b respectively.

So,

$$3a + b = 400 \dots \dots \dots \text{(I)}$$

$$2a + b = 300 \dots \dots \dots \text{(II)}$$

By solving equations (I) and (II), we get,

$$6a + 2b = 800$$

$$6a + 3b = 900$$

$$b = 100$$

Now, put the value of b in equation (I), we get,

$$3a + b = 400$$

$$3a + 100 = 400$$

$$3a = 300$$

$$a = 100$$

Red Fort

The total number of males (Indian + Foreigners) who visited = 450

The total number of females (Indian + Foreigners) who visited = $\{(450/180)*100\} = 250$

The total number of visitors who visited = $(450+250) = 700$

Now, let the number of Indian males and females who visited be 5c and 3c respectively.

And, let the number of foreigner males and females who visited be 2d and d respectively.

So,

$$5c + 2d = 450 \dots \dots \dots (III)$$

$$3c + d = 250 \dots \dots \dots (IV)$$

By solving equation (III) and (IV), we get,

$$15c + 6d = 1350$$

$$15c + 5d = 1250$$

$$d = 100$$

By solving equation (III) and (IV), we get,

$$5c + 2d = 450$$

$$5c + (2*100) = 450$$

$$5c = 450 - 200$$

$$5c = 250$$

$$c = 50$$

India Gate

The total number of visitors who visited = 500

The total number of females (Indian + Foreigners) who visited = $(400-190) = 210$

The total number of males (Indian + Foreigners) who visited = $(500-210) = 290$

Now, let the number of Indian males and females who visited be 4e and 3e respectively.

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And, let the number of foreigner males and females who visited be $3f$ and $2f$ respectively.

So,

$$4e + 3f = 290 \dots\dots\dots(V)$$

$$3e + 2f = 210 \dots\dots\dots(VI)$$

By solving equation (V) and (VI), we get,

$$12e + 9f = 870$$

$$12e + 8f = 840$$

$$f = 30$$

Now, put the value of f in equation (V), we get,

$$4e + (3 \times 30) = 290$$

$$4e = 290 - 90$$

$$4e = 200$$

$$e = 50$$

Hampi

The total number of visitors who visited = $(500+550) = 1050$

The total number of females (Indian + Foreigners) who visited = 400

The total number of males (Indian + Foreigners) who visited = $(1050-400) = 650$

Now, let the number of Indian males and females who visited be $2g$ and g respectively.

And, let the number of foreigner males and females who visited be $5h$ and $4h$ respectively.

So,

$$2g + 5h = 650 \dots\dots\dots(VII)$$

$$g + 4h = 400 \dots\dots\dots(VIII)$$

By solving equation (VII) and (VIII), we get,

$$2g + 5h = 650$$

$$2g + 8h = 800$$

$$3h = 150$$

$$h = 50$$

By solving equation (VII) and (VIII), we get,

$$2g + 5h = 650$$

$$2g + (5 \times 50) = 650$$

$$2g = 650 - 250$$

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$$2g = 400$$

$$g = 200$$

Taj Mahal

The number of Indian males who visited = $3a = (3 \times 100) = 300$

The number of Indian females who visited = $2a = (2 \times 100) = 200$

The number of foreigner males who visited = $b = 100$

The number of foreigner females who visited = $b = 100$

So, the total number of Indians who visited = $(300+200) = 500$

And, the total number of foreigners who visited = $(100+100) = 200$

Red Fort

The number of Indian males who visited = $5c = (5 \times 50) = 250$

The number of Indian females who visited = $3c = (3 \times 50) = 150$

The number of foreigner males who visited = $2d = (2 \times 100) = 200$

The number of foreigner females who visited = $d = 100$

So, the total number of Indians who visited = $(250+150) = 400$

And, the total number of foreigners who visited = $(200+100) = 300$

India Gate

The number of Indian males who visited = $4e = (4 \times 50) = 200$

The number of Indian females who visited = $3e = (3 \times 50) = 150$

The number of foreigner males who visited = $3f = (3 \times 30) = 90$

The number of foreigner females who visited = $2f = (2 \times 30) = 60$

So, the total number of Indians who visited = $(200+150) = 350$

And, the total number of foreigners who visited = $(90+60) = 150$

Hampi

The number of Indian males who visited = $2g = (2 \times 200) = 400$

The number of Indian females who visited = $g = 200$

The number of foreigner males who visited = $5h = (5 \times 50) = 250$

The number of foreigner females who visited = $4h = (4 \times 50) = 200$

So, the total number of Indians who visited = $(400+200) = 600$

And, the total number of foreigners who visited = $(250+200) = 450$

Indians

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Name of the place	The total number of Indian visitors who visited	The number of Indian males who visited	The number of Indian females who visited
Taj Mahal	500	300	200
Red Fort	400	250	150
India Gate	350	200	150
Hampi	600	400	200
Total	1850	1150	700

Foreigners

Name of the place	The total number of foreigner visitors who visited	The number of foreigner males who visited	The number of foreigner females who visited
Taj Mahal	200	100	100
Red Fort	300	200	100
India Gate	150	90	60
Hampi	450	250	200
Total	1100	640	460

Answer: D

Hampi

The total number of indian males who visited = 400

The total number of Indian females who visited = 200

Now,

$$(400-X)/(200+2X) = 7/6$$

$$2400 - 6X = 1400 + 14X$$

$$20X = 1000$$

$$X = 50$$

50 is exactly divisible by 10.

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Hence, the correct answer is option D.

11. Questions

Shop X

The M.P. of the digital watch = Rs. 1800

The S.P. of the digital watch = $\frac{8}{9}$ of 1800 = Rs. 1600 (**Note: 11.11% = $\frac{1}{9}$**)

Let the C.P. of the smart watch be Rs. 100a

So, the M.P. of the smart watch = 140% of 100a = Rs. 140a

And, the S.P. of the smart watch = 120% of 100a = Rs. 120a

Now, the C.P. of digital watch = Rs. 120a

The overall M.P. of digital and smart watch together = Rs. (1800+140a)

The overall S.P. of digital and smart watch together = Rs. (1600+120a)

So,

$$(1800+140a) - (1600+120a) = 400$$

$$1400 + 140a = 1600 + 120a$$

$$20a = 200$$

$$a = 10$$

Shop Y

The C.P. of the digital watch = Rs. 2000

The S.P. of the digital watch = 150% of 2000 = Rs. 3000

The S.P. of the smart watch = Rs. 2500

The M.P. of the smart watch = $\{(2500/5)*6\}$ = Rs. 3000 (**Note: 16.67% = $\frac{1}{6}$**)

Let the M.P. of the digital watch and the C.P. of the smart watch be Rs. 32b and Rs. 15b respectively.

The mark-up amount of the digital watch = (32b-2000)

The mark-up amount of the smart watch = (3000-15b)

Now,

$$(3000-15b) - (32b-2000) = 300$$

$$2700 - 15b = 32b - 2000$$

$$47b = 4700$$

$$b = 100$$

Shop Z

Let the C.P. of the digital watch be Rs. 100c

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So, the M.P. of the digital watch = 125% of 100c = Rs. 125c

The S.P. of the digital watch = Rs. 2700

The C.P. of the smart watch = Rs. 1800

The M.P. of the smart watch = Rs. (125c-840)

The S.P. of the smart watch = {(125c-840)-160} = Rs. (125c-1000)

Now,

$$100c/(125c-1000) = 6/5$$

$$500c = 750c - 6000$$

$$250c = 6000$$

$$c = 24$$

Shop X

The C.P. of the digital watch = 120a = 1200

The C.P. of the smart watch = 100a = (100*10) = Rs. 1000

The M.P. of the smart watch = 140a = (140*10) = Rs. 1400

The S.P. of the smart watch = 120a = (120*10) = Rs. 1200

Shop Y

The M.P. of the digital watch = 32b = (32*100) = Rs. 3200

The C.P. of the smart watch = 15b = (15*100) = Rs. 1500

Shop Z

The C.P. of the digital watch = 100c = (100*24) = Rs. 2400

So, the M.P. of the digital watch = 125c = (125*24) = Rs. 3000

The M.P. of the smart watch = (125c-840) = (3000-840) = Rs. 2160

The S.P. of the smart watch = (125c-1000) = (3000-1000) = Rs. 2000

Digital Watch

Shop	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
X	1200	1800	1600
Y	2000	3200	3000
Z	2400	3000	2700
Total	5600	8000	7300

Smart Watch

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Shop	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
X	1000	1400	1200
Y	1500	3000	2500
Z	1800	2160	2000
Total	4300	6560	5700

Answer: A

Now, according to the question,

Shop X (digital watch)

The C.P. = Rs. 1200

The M.P. = Rs. 1800

Using I

The S.P. = Rs. (1600-100) = Rs. 1500

Now, the profit percentage = $\left[\frac{(1500-1200)}{1200} \times 100\right]\% = \left\{\frac{(300)}{1200} \times 100\right\}\% = 25\%$.

So, I does satisfy.

Using II

The S.P. = Rs. (1600-200) = Rs. 1400

Now, the profit percentage = $\left[\frac{(1400-1200)}{1200} \times 100\right]\% = \left\{\frac{(200)}{1200} \times 100\right\}\% = 16.67\%$.

So, II does not satisfy.

Using III

The S.P. = Rs. (1600-300) = Rs. 1300

Now, the profit percentage = $\left[\frac{(1300-1200)}{1200} \times 100\right]\% = \left\{\frac{(100)}{1200} \times 100\right\}\% = 8.33\%$.

So, III does not satisfy.

Therefore, only I follow.

Hence, the correct answer is option A.

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12. Questions

Shop X

The M.P. of the digital watch = Rs. 1800

The S.P. of the digital watch = $\frac{8}{9}$ of 1800 = Rs. 1600 (**Note: $11.11\% = \frac{1}{9}$**)

Let the C.P. of the smart watch be Rs. 100a

So, the M.P. of the smart watch = 140% of 100a = Rs. 140a

And, the S.P. of the smart watch = 120% of 100a = Rs. 120a

Now, the C.P. of digital watch = Rs. 120a

The overall M.P. of digital and smart watch together = Rs. (1800+140a)

The overall S.P. of digital and smart watch together = Rs. (1600+120a)

So,

$$(1800+140a) - (1600+120a) = 400$$

$$1400 + 140a = 1600 + 120a$$

$$20a = 200$$

$$a = 10$$

Shop Y

The C.P. of the digital watch = Rs. 2000

The S.P. of the digital watch = 150% of 2000 = Rs. 3000

The S.P. of the smart watch = Rs. 2500

The M.P. of the smart watch = $\{(2500/5)*6\}$ = Rs. 3000 (**Note: 16.67% = 1/6**)

Let the M.P. of the digital watch and the C.P. of the smart watch be Rs. 32b and Rs. 15b respectively.

The mark-up amount of the digital watch = (32b-2000)

The mark-up amount of the smart watch = (3000-15b)

Now,

$$(3000-15b) - (32b-2000) = 300$$

$$2700 - 15b = 32b - 2000$$

$$47b = 4700$$

$$b = 100$$

Shop Z

Let the C.P. of the digital watch be Rs. 100c

So, the M.P. of the digital watch = 125% of 100c = Rs. 125c

The S.P. of the digital watch = Rs. 2700

The C.P. of the smart watch = Rs. 1800

The M.P. of the smart watch = Rs. (125c-840)

The S.P. of the smart watch = $\{(125c-840)-160\}$ = Rs. (125c-1000)

Now,

$$100c/(125c-1000) = 6/5$$

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$$500c = 750c - 6000$$

$$250c = 6000$$

$$c = 24$$

Shop X

The C.P. of the digital watch = $120a = 1200$

The C.P. of the smart watch = $100a = (100 \times 10) = \text{Rs. } 1000$

The M.P. of the smart watch = $140a = (140 \times 10) = \text{Rs. } 1400$

The S.P. of the smart watch = $120a = (120 \times 10) = \text{Rs. } 1200$

Shop Y

The M.P. of the digital watch = $32b = (32 \times 100) = \text{Rs. } 3200$

The C.P. of the smart watch = $15b = (15 \times 100) = \text{Rs. } 1500$

Shop Z

The C.P. of the digital watch = $100c = (100 \times 24) = \text{Rs. } 2400$

So, the M.P. of the digital watch = $125c = (125 \times 24) = \text{Rs. } 3000$

The M.P. of the smart watch = $(125c - 840) = (3000 - 840) = \text{Rs. } 2160$

The S.P. of the smart watch = $(125c - 1000) = (3000 - 1000) = \text{Rs. } 2000$

Digital Watch

Shop	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
X	1200	1800	1600
Y	2000	3200	3000
Z	2400	3000	2700
Total	5600	8000	7300

Smart Watch

Shop	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
X	1000	1400	1200
Y	1500	3000	2500
Z	1800	2160	2000
Total	4300	6560	5700

Answer: B

Shop Y (smart watch)

The C.P. of each watch = Rs. 1500

The S.P. of each watch = Rs. 2500

The total number of watches unsold = 4

Let the number of smart watches purchased be n .

Therefore, the total number of smart watches sold = $(n-4)$

So, the overall C.P. of all watches purchased = $(n \times 1500) = \text{Rs. } 1500n$

The overall S.P. of all watches sold = $\{(n-4) \times 2500\} = (2500n - 10000)$

Now,

$$(2500n - 10000) - 1500n = 2000$$

$$2500n - 12000 = 1500n$$

$$1000n = 12000$$

$$n = 12$$

Therefore, the total number of smart watches purchased by shop Y = $n = 12$.

Hence, the correct answer is option B.

13. Questions

Shop X

The M.P. of the digital watch = Rs. 1800

The S.P. of the digital watch = $\frac{8}{9}$ of 1800 = Rs. 1600 (**Note: 11.11% = $\frac{1}{9}$**)

Let the C.P. of the smart watch be Rs. $100a$

So, the M.P. of the smart watch = 140% of $100a = \text{Rs. } 140a$

And, the S.P. of the smart watch = 120% of $100a = \text{Rs. } 120a$

Now, the C.P. of digital watch = Rs. $120a$

The overall M.P. of digital and smart watch together = Rs. $(1800 + 140a)$

The overall S.P. of digital and smart watch together = Rs. $(1600 + 120a)$

So,

$$(1800 + 140a) - (1600 + 120a) = 400$$

$$1400 + 140a = 1600 + 120a$$

$$20a = 200$$

$$a = 10$$

Shop Y

The C.P. of the digital watch = Rs. 2000

The S.P. of the digital watch = 150% of 2000 = Rs. 3000

The S.P. of the smart watch = Rs. 2500

The M.P. of the smart watch = $\{(2500/5)*6\}$ = Rs. 3000 (**Note: 16.67% = 1/6**)

Let the M.P. of the digital watch and the C.P. of the smart watch be Rs. 32b and Rs. 15b respectively.

The mark-up amount of the digital watch = (32b-2000)

The mark-up amount of the smart watch = (3000-15b)

Now,

$$(3000-15b) - (32b-2000) = 300$$

$$2700 - 15b = 32b - 2000$$

$$47b = 4700$$

$$b = 100$$

Shop Z

Let the C.P. of the digital watch be Rs. 100c

So, the M.P. of the digital watch = 125% of 100c = Rs. 125c

The S.P. of the digital watch = Rs. 2700

The C.P. of the smart watch = Rs. 1800

The M.P. of the smart watch = Rs. (125c-840)

The S.P. of the smart watch = $\{(125c-840)-160\}$ = Rs. (125c-1000)

Now,

$$100c/(125c-1000) = 6/5$$

$$500c = 750c - 6000$$

$$250c = 6000$$

$$c = 24$$

Shop X

The C.P. of the digital watch = 120a = 1200

The C.P. of the smart watch = 100a = (100*10) = Rs. 1000

The M.P. of the smart watch = 140a = (140*10) = Rs. 1400

The S.P. of the smart watch = 120a = (120*10) = Rs. 1200

Shop Y

The M.P. of the digital watch = 32b = (32*100) = Rs. 3200

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The C.P. of the smart watch = $15b = (15 \times 100) = \text{Rs. } 1500$

Shop Z

The C.P. of the digital watch = $100c = (100 \times 24) = \text{Rs. } 2400$

So, the M.P. of the digital watch = $125c = (125 \times 24) = \text{Rs. } 3000$

The M.P. of the smart watch = $(125c - 840) = (3000 - 840) = \text{Rs. } 2160$

The S.P. of the smart watch = $(125c - 1000) = (3000 - 1000) = \text{Rs. } 2000$

Digital Watch

Shop	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
X	1200	1800	1600
Y	2000	3200	3000
Z	2400	3000	2700
Total	5600	8000	7300

Smart Watch

Shop	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
X	1000	1400	1200
Y	1500	3000	2500
Z	1800	2160	2000
Total	4300	6560	5700

Answer: D

Shop W

Digital Watch

The C.P. = Rs. 800

The M.P. = $\{800 + (M\% \text{ of } 800)\} = \text{Rs. } (800 + 8M)$

The S.P. = $126\% \text{ of } 800 = 1008$

Now,

$$(800 + 8M) - 1008 = 112$$

$$800 + 8M = 1120$$

$$8M = 320$$

$$M = 40$$

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So, the C.P. of the smart watch = $(16M+60) = (16 \times 40 + 60) = (640 + 60) = \text{Rs. } 700$

Therefore, the S.P. of the smart watch in shop W = 110% of 700 = Rs. 770.

Hence, the correct answer is option D.

14. Questions

Shop X

The M.P. of the digital watch = Rs. 1800

The S.P. of the digital watch = $\frac{8}{9}$ of 1800 = Rs. 1600 (**Note: 11.11% = $\frac{1}{9}$**)

Let the C.P. of the smart watch be Rs. 100a

So, the M.P. of the smart watch = 140% of 100a = Rs. 140a

And, the S.P. of the smart watch = 120% of 100a = Rs. 120a

Now, the C.P. of digital watch = Rs. 120a

The overall M.P. of digital and smart watch together = Rs. (1800+140a)

The overall S.P. of digital and smart watch together = Rs. (1600+120a)

So,

$$(1800+140a) - (1600+120a) = 400$$

$$1400 + 140a = 1600 + 120a$$

$$20a = 200$$

$$a = 10$$

Shop Y

The C.P. of the digital watch = Rs. 2000

The S.P. of the digital watch = 150% of 2000 = Rs. 3000

The S.P. of the smart watch = Rs. 2500

The M.P. of the smart watch = $\{(2500/5) \times 6\} = \text{Rs. } 3000$ (**Note: 16.67% = $\frac{1}{6}$**)

Let the M.P. of the digital watch and the C.P. of the smart watch be Rs. 32b and Rs. 15b respectively.

The mark-up amount of the digital watch = (32b-2000)

The mark-up amount of the smart watch = (3000-15b)

Now,

$$(3000-15b) - (32b-2000) = 300$$

$$2700 - 15b = 32b - 2000$$

$$47b = 4700$$

$$b = 100$$

Shop Z

Let the C.P. of the digital watch be Rs. $100c$

So, the M.P. of the digital watch = 125% of $100c$ = Rs. $125c$

The S.P. of the digital watch = Rs. 2700

The C.P. of the smart watch = Rs. 1800

The M.P. of the smart watch = Rs. $(125c-840)$

The S.P. of the smart watch = $\{(125c-840)-160\}$ = Rs. $(125c-1000)$

Now,

$$100c/(125c-1000) = 6/5$$

$$500c = 750c - 6000$$

$$250c = 6000$$

$$c = 24$$

Shop X

The C.P. of the digital watch = $120a$ = 1200

The C.P. of the smart watch = $100a$ = $(100*10)$ = Rs. 1000

The M.P. of the smart watch = $140a$ = $(140*10)$ = Rs. 1400

The S.P. of the smart watch = $120a$ = $(120*10)$ = Rs. 1200

Shop Y

The M.P. of the digital watch = $32b$ = $(32*100)$ = Rs. 3200

The C.P. of the smart watch = $15b$ = $(15*100)$ = Rs. 1500

Shop Z

The C.P. of the digital watch = $100c$ = $(100*24)$ = Rs. 2400

So, the M.P. of the digital watch = $125c$ = $(125*24)$ = Rs. 3000

The M.P. of the smart watch = $(125c-840)$ = $(3000-840)$ = Rs. 2160

The S.P. of the smart watch = $(125c-1000)$ = $(3000-1000)$ = Rs. 2000

Digital Watch

Shop	The costprice (in Rs.)	The markedprice (in Rs.)	The sellingprice (in Rs.)
X	1200	1800	1600
Y	2000	3200	3000
Z	2400	3000	2700
Total	5600	8000	7300

Smart Watch

Shop	The costprice (in Rs.)	The markedprice (in Rs.)	The sellingprice (in Rs.)
X	1000	1400	1200
Y	1500	3000	2500
Z	1800	2160	2000
Total	4300	6560	5700

Answer: A

Shop Z

Analogue Watch

The M.P. = Rs. K

The S.P. = Rs. 700

The C.P. = $(700 - 200) = \text{Rs. } 500$

Now,

$K = 160\% \text{ of } 500$

$K = 800$

So, the M.P. of the analog watch = $K = \text{Rs. } 800$

The M.P. of the digital watch = Rs. 3000

Therefore, the required difference = $(3000 - 800) = \text{Rs. } 2200$.

Hence, the correct answer is option A.

15. Questions

Shop X

The M.P. of the digital watch = Rs. 1800

The S.P. of the digital watch = $\frac{8}{9}$ of 1800 = Rs. 1600 (**Note: $11.11\% = \frac{1}{9}$**)

Let the C.P. of the smart watch be Rs. 100a

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So, the M.P. of the smart watch = 140% of 100a = Rs. 140a

And, the S.P. of the smart watch = 120% of 100a = Rs. 120a

Now, the C.P. of digital watch = Rs. 120a

The overall M.P. of digital and smart watch together = Rs. (1800+140a)

The overall S.P. of digital and smart watch together = Rs. (1600+120a)

So,

$$(1800+140a) - (1600+120a) = 400$$

$$1400 + 140a = 1600 + 120a$$

$$20a = 200$$

$$a = 10$$

Shop Y

The C.P. of the digital watch = Rs. 2000

The S.P. of the digital watch = 150% of 2000 = Rs. 3000

The S.P. of the smart watch = Rs. 2500

The M.P. of the smart watch = $\{(2500/5)*6\}$ = Rs. 3000 (**Note: 16.67% = 1/6**)

Let the M.P. of the digital watch and the C.P. of the smart watch be Rs. 32b and Rs. 15b respectively.

The mark-up amount of the digital watch = (32b-2000)

The mark-up amount of the smart watch = (3000-15b)

Now,

$$(3000-15b) - (32b-2000) = 300$$

$$2700 - 15b = 32b - 2000$$

$$47b = 4700$$

$$b = 100$$

Shop Z

Let the C.P. of the digital watch be Rs. 100c

So, the M.P. of the digital watch = 125% of 100c = Rs. 125c

The S.P. of the digital watch = Rs. 2700

The C.P. of the smart watch = Rs. 1800

The M.P. of the smart watch = Rs. (125c-840)

The S.P. of the smart watch = $\{(125c-840)-160\}$ = Rs. (125c-1000)

Now,

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$$100c/(125c-1000) = 6/5$$

$$500c = 750c - 6000$$

$$250c = 6000$$

$$c = 24$$

Shop X

The C.P. of the digital watch = $120a = 1200$

The C.P. of the smart watch = $100a = (100 \times 10) = \text{Rs. } 1000$

The M.P. of the smart watch = $140a = (140 \times 10) = \text{Rs. } 1400$

The S.P. of the smart watch = $120a = (120 \times 10) = \text{Rs. } 1200$

Shop Y

The M.P. of the digital watch = $32b = (32 \times 100) = \text{Rs. } 3200$

The C.P. of the smart watch = $15b = (15 \times 100) = \text{Rs. } 1500$

Shop Z

The C.P. of the digital watch = $100c = (100 \times 24) = \text{Rs. } 2400$

So, the M.P. of the digital watch = $125c = (125 \times 24) = \text{Rs. } 3000$

The M.P. of the smart watch = $(125c - 840) = (3000 - 840) = \text{Rs. } 2160$

The S.P. of the smart watch = $(125c - 1000) = (3000 - 1000) = \text{Rs. } 2000$

Digital Watch

Shop	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
X	1200	1800	1600
Y	2000	3200	3000
Z	2400	3000	2700
Total	5600	8000	7300

Smart Watch

Shop	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
X	1000	1400	1200
Y	1500	3000	2500
Z	1800	2160	2000
Total	4300	6560	5700

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Answer: C

Shop Z (smart watch)

The C.P. = Rs. 1800

The M.P. = Rs. L

The S.P. = Rs. 2000

Now,

$$L - 2000 = 400$$

$$L = 2400$$

So, the value of $\sqrt{(L+100)} = \sqrt{(2400+100)} = \sqrt{2500} = 50$.

Hence, the correct answer is option C.

16. Questions

Let the total number of questions contained in subject P and Q 300m and 500m respectively.

Subject P

The total number of questions contained = 300m

The number of questions attempted = 75% of 300m = 225m

The number of questions skipped = 25% of 300m = 75m

The number of correct attempts = 60% of 225m = 135m

The number of wrong attempts = 40% of 225m = 90m

Subject Q

The total number of questions contained = 500m

The number of questions attempted = 80% of 500m = 400m

The number of questions skipped = 20% of 500m = 100m

The number of correct attempts = 90% of 400m = 360m

The number of wrong attempts = 10% of 400m = 40m

Now,

$$90m - 40m = 20$$

$$50m = 20$$

$$m = 0.40$$

Also, the ratio the of the total number of questions contained in subject R and S = 1:2. Let it be 100n and 200n respectively.

Subject R

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The total number of questions contained = $100n$

The number of questions attempted = 90% of $100n = 90n$

The number of questions skipped = 10% of $100n = 10n$

The number of correct attempts = 75% of $90n = 67.50n$

The number of wrong attempts = 25% of $90n = 22.50n$

Subject S

The total number of questions contained = $200n$

The number of questions attempted = 70% of $200n = 140n$

The number of questions skipped = 30% of $200n = 60n$

The number of correct attempts = 50% of $140n = 70n$

The number of wrong attempts = 50% of $140n = 70n$

Now,

$$70n - 67.50n = 2$$

$$2.50n = 2$$

$$n = 0.80$$

Subject P

The total number of questions contained = $300m = (300 \times 0.40) = 120$

The number of questions attempted = $225m = (225 \times 0.40) = 90$

The number of questions skipped = $75m = (75 \times 0.40) = 30$

The number of correct attempts = $135m = (135 \times 0.40) = 54$

The number of wrong attempts = $90m = (90 \times 0.40) = 36$

Subject Q

The total number of questions contained = $500m = (500 \times 0.40) = 200$

The number of questions attempted = $400m = (400 \times 0.40) = 160$

The number of questions skipped = $100m = (100 \times 0.40) = 40$

The number of correct attempts = $360m = (360 \times 0.40) = 144$

The number of wrong attempts = $40m = (40 \times 0.40) = 16$

Subject R

The total number of questions contained = $100n = (100 \times 0.80) = 80$

The number of questions attempted = $90n = (90 \times 0.80) = 72$

The number of questions skipped = $10n = (10 \times 0.80) = 8$

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The number of correct attempts = $67.50n = (67.50 \times 0.80) = 54$

The number of wrong attempts = $22.50n = (22.50 \times 0.80) = 18$

Subject S

The total number of questions contained = $200n = (200 \times 0.80) = 160$

The number of questions attempted = $140n = (140 \times 0.80) = 112$

The number of questions skipped = $60n = (60 \times 0.80) = 48$

The number of correct attempts = $70n = (70 \times 0.80) = 56$

The number of wrong attempts = $70n = 56$

Subject T

The number of questions skipped = $(48+60) = 108$

The number of questions attempted = $\{(108/45 \times 55)\} = 132$

The total number of questions contained = $(108+132) = 240$

The number of correct attempts = $75\% \text{ of } 132 = 99$

The number of wrong attempts = $25\% \text{ of } 132 = 33$

Subject	The total number of questions contained	The number of questions attempted	The number of questions skipped	The number of correct attempts	The number of wrong attempts
P	120	90	30	54	36
Q	200	160	40	144	16
R	80	72	8	54	18
S	160	112	48	56	56
T	240	132	108	99	33
Total	800	566	234	407	159

Answer: E

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Now, according to the question,

Subject P

The marks deducted for each wrong attempt = A

So, the marks awarded for each correct attempt = $(A \times 4) = 4A$

The number of correct attempts = 54

The number of wrong attempts = 36

The total marks obtained = $\{(54 \times 4A) - (36 \times A)\} = (216A - 36A) = 180A$

Now,

$$180A = 45$$

$$A = 0.25$$

I). 0.25 is not an integer.

So, I is definitely not true.

II). 0.25 is not a whole number.

So, II is definitely not true.

III). 0.25 is not a prime number.

So, II is definitely not true.

Therefore, none is true.

Hence, the correct answer is option E.

17. Questions

Let the total number of questions contained in subject P and Q 300m and 500m respectively.

Subject P

The total number of questions contained = 300m

The number of questions attempted = 75% of 300m = 225m

The number of questions skipped = 25% of 300m = 75m

The number of correct attempts = 60% of 225m = 135m

The number of wrong attempts = 40% of 225m = 90m

Subject Q

The total number of questions contained = 500m

The number of questions attempted = 80% of 500m = 400m

The number of questions skipped = 20% of 500m = 100m

The number of correct attempts = 90% of 400m = 360m

The number of wrong attempts = 10% of 400m = 40m

Now,

$$90m - 40m = 20$$

$$50m = 20$$

$$m = 0.40$$

Also, the ratio the of the total number of questions contained in subject R and S = 1:2. Let it be 100n and 200n respectively.

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Subject R

The total number of questions contained = $100n$

The number of questions attempted = 90% of $100n = 90n$

The number of questions skipped = 10% of $100n = 10n$

The number of correct attempts = 75% of $90n = 67.50n$

The number of wrong attempts = 25% of $90n = 22.50n$

Subject S

The total number of questions contained = $200n$

The number of questions attempted = 70% of $200n = 140n$

The number of questions skipped = 30% of $200n = 60n$

The number of correct attempts = 50% of $140n = 70n$

The number of wrong attempts = 50% of $140n = 70n$

Now,

$$70n - 67.50n = 2$$

$$2.50n = 2$$

$$n = 0.80$$

Subject P

The total number of questions contained = $300m = (300 \times 0.40) = 120$

The number of questions attempted = $225m = (225 \times 0.40) = 90$

The number of questions skipped = $75m = (75 \times 0.40) = 30$

The number of correct attempts = $135m = (135 \times 0.40) = 54$

The number of wrong attempts = $90m = (90 \times 0.40) = 36$

Subject Q

The total number of questions contained = $500m = (500 \times 0.40) = 200$

The number of questions attempted = $400m = (400 \times 0.40) = 160$

The number of questions skipped = $100m = (100 \times 0.40) = 40$

The number of correct attempts = $360m = (360 \times 0.40) = 144$

The number of wrong attempts = $40m = (40 \times 0.40) = 16$

Subject R

The total number of questions contained = $100n = (100 \times 0.80) = 80$

The number of questions attempted = $90n = (90 \times 0.80) = 72$

The number of questions skipped = $10n = (10 \times 0.80) = 8$

The number of correct attempts = $67.50n = (67.50 \times 0.80) = 54$

The number of wrong attempts = $22.50n = (22.50 \times 0.80) = 18$

Subject S

The total number of questions contained = $200n = (200 \times 0.80) = 160$

The number of questions attempted = $140n = (140 \times 0.80) = 112$

The number of questions skipped = $60n = (60 \times 0.80) = 48$

The number of correct attempts = $70n = (70 \times 0.80) = 56$

The number of wrong attempts = $70n = 56$

Subject T

The number of questions skipped = $(48+60) = 108$

The number of questions attempted = $\{(108/45 \times 55)\} = 132$

The total number of questions contained = $(108+132) = 240$

The number of correct attempts = $75\% \text{ of } 132 = 99$

The number of wrong attempts = $25\% \text{ of } 132 = 33$

Subject	The total number of questions contained	The number of questions attempted	The number of questions skipped	The number of correct attempts	The number of wrong attempts
P	120	90	30	54	36
Q	200	160	40	144	16
R	80	72	8	54	18
S	160	112	48	56	56
T	240	132	108	99	33
Total	800	566	234	407	159

Answer: B

Subject Q

The marks awarded for each correct attempt in subjective section = 1

The marks awarded for each correct attempt in objective section = $(1/2) = 0.50$

The marks deducted for each wrong attempt in objective section = 0.10

The marks deducted for each wrong attempt in subjective section = $\{(0.10/2) \times 5\} = 0.25$

The total number of correct attempts (subjective + objective) = 144

The number of correct attempts in objective section = $1/12$ of 144 = 12 (**Note: 8.33% = 1/12**)

The number of correct attempts in subjective section = $(144-12) = 132$

The total number of wrong attempts (subjective + objective) = 16

The number of wrong attempts in objective section = $1/4$ of 16 = 4

The number of wrong attempts in subjective section = $(16-4) = 12$

So, the marks obtained in subjective section = $\{(132*1)-(12*0.25)\} = (132-3) = 129$

So, the marks obtained in objective section = $\{(12*0.50)-(4*0.10)\} = (6-0.40) = 5.60$

Therefore, the total marks obtained by the student in subject Q = $(129+5.60) = 134.60$.

Hence, the correct answer is option B.

18. Questions

Let the total number of questions contained in subject P and Q 300m and 500m respectively.

Subject P

The total number of questions contained = 300m

The number of questions attempted = 75% of 300m = 225m

The number of questions skipped = 25% of 300m = 75m

The number of correct attempts = 60% of 225m = 135m

The number of wrong attempts = 40% of 225m = 90m

Subject Q

The total number of questions contained = 500m

The number of questions attempted = 80% of 500m = 400m

The number of questions skipped = 20% of 500m = 100m

The number of correct attempts = 90% of 400m = 360m

The number of wrong attempts = 10% of 400m = 40m

Now,

$$90m - 40m = 20$$

$$50m = 20$$

$$m = 0.40$$

Also, the ratio the of the total number of questions contained in subject R and S = 1:2. Let it be 100n and 200n respectively.

Subject R

The total number of questions contained = 100n

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The number of questions attempted = 90% of 100n = 90n

The number of questions skipped = 10% of 100n = 10n

The number of correct attempts = 75% of 90n = 67.50n

The number of wrong attempts = 25% of 90n = 22.50n

Subject S

The total number of questions contained = 200n

The number of questions attempted = 70% of 200n = 140n

The number of questions skipped = 30% of 200n = 60n

The number of correct attempts = 50% of 140n = 70n

The number of wrong attempts = 50% of 140n = 70n

Now,

$$70n - 67.50n = 2$$

$$2.50n = 2$$

$$n = 0.80$$

Subject P

The total number of questions contained = 300m = (300*0.40) = 120

The number of questions attempted = 225m = (225*0.40) = 90

The number of questions skipped = 75m = (75*0.40) = 30

The number of correct attempts = 135m = (135*0.40) = 54

The number of wrong attempts = 90m = (90*0.40) = 36

Subject Q

The total number of questions contained = 500m = (500*0.40) = 200

The number of questions attempted = 400m = (400*0.40) = 160

The number of questions skipped = 100m = (100*0.40) = 40

The number of correct attempts = 360m = (360*0.40) = 144

The number of wrong attempts = 40m = (40*0.40) = 16

Subject R

The total number of questions contained = 100n = (100*0.80) = 80

The number of questions attempted = 90n = (90*0.80) = 72

The number of questions skipped = 10n = (10*0.80) = 8

The number of correct attempts = 67.50n = (67.50*0.80) = 54

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The number of wrong attempts = $22.50n = (22.50 \times 0.80) = 18$

Subject S

The total number of questions contained = $200n = (200 \times 0.80) = 160$

The number of questions attempted = $140n = (140 \times 0.80) = 112$

The number of questions skipped = $60n = (60 \times 0.80) = 48$

The number of correct attempts = $70n = (70 \times 0.80) = 56$

The number of wrong attempts = $70n = 56$

Subject T

The number of questions skipped = $(48+60) = 108$

The number of questions attempted = $\{(108/45 \times 55)\} = 132$

The total number of questions contained = $(108+132) = 240$

The number of correct attempts = $75\% \text{ of } 132 = 99$

The number of wrong attempts = $25\% \text{ of } 132 = 33$

Subject	The total number of questions contained	The number of questions attempted	The number of questions skipped	The number of correct attempts	The number of wrong attempts
P	120	90	30	54	36
Q	200	160	40	144	16
R	80	72	8	54	18
S	160	112	48	56	56
T	240	132	108	99	33
Total	800	566	234	407	159

Answer: A

Subject S

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The total number of questions attempted = 112

The total number of questions skipped = 48

The number of questions skipped in practical section = $1/4 \text{ of } 48 = 12$

The number of questions skipped in theory section = $(48-12) = 36$

The number of questions attempted in theory section = $(36+28) = 64$

So, $X = 64$

The number of questions attempted in practical section = $(112-64) = 48$

So, $Y = 48$

Therefore, $X > Y$.

Hence, the correct answer is option A.

19. Questions

Let the total number of questions contained in subject P and Q 300m and 500m respectively.

Subject P

The total number of questions contained = 300m

The number of questions attempted = 75% of 300m = 225m

The number of questions skipped = 25% of 300m = 75m

The number of correct attempts = 60% of 225m = 135m

The number of wrong attempts = 40% of 225m = 90m

Subject Q

The total number of questions contained = 500m

The number of questions attempted = 80% of 500m = 400m

The number of questions skipped = 20% of 500m = 100m

The number of correct attempts = 90% of 400m = 360m

The number of wrong attempts = 10% of 400m = 40m

Now,

$$90m - 40m = 20$$

$$50m = 20$$

$$m = 0.40$$

Also, the ratio the of the total number of questions contained in subject R and S = 1:2. Let it be 100n and 200n respectively.

Subject R

The total number of questions contained = 100n

The number of questions attempted = 90% of 100n = 90n

The number of questions skipped = 10% of 100n = 10n

The number of correct attempts = 75% of 90n = 67.50n

The number of wrong attempts = 25% of 90n = 22.50n

Subject S

The total number of questions contained = 200n

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The number of questions attempted = 70% of 200n = 140n

The number of questions skipped = 30% of 200n = 60n

The number of correct attempts = 50% of 140n = 70n

The number of wrong attempts = 50% of 140n = 70n

Now,

$$70n - 67.50n = 2$$

$$2.50n = 2$$

$$n = 0.80$$

Subject P

The total number of questions contained = 300m = (300*0.40) = 120

The number of questions attempted = 225m = (225*0.40) = 90

The number of questions skipped = 75m = (75*0.40) = 30

The number of correct attempts = 135m = (135*0.40) = 54

The number of wrong attempts = 90m = (90*0.40) = 36

Subject Q

The total number of questions contained = 500m = (500*0.40) = 200

The number of questions attempted = 400m = (400*0.40) = 160

The number of questions skipped = 100m = (100*0.40) = 40

The number of correct attempts = 360m = (360*0.40) = 144

The number of wrong attempts = 40m = (40*0.40) = 16

Subject R

The total number of questions contained = 100n = (100*0.80) = 80

The number of questions attempted = 90n = (90*0.80) = 72

The number of questions skipped = 10n = (10*0.80) = 8

The number of correct attempts = 67.50n = (67.50*0.80) = 54

The number of wrong attempts = 22.50n = (22.50*0.80) = 18

Subject S

The total number of questions contained = 200n = (200*0.80) = 160

The number of questions attempted = 140n = (140*0.80) = 112

The number of questions skipped = 60n = (60*0.80) = 48

The number of correct attempts = 70n = (70*0.80) = 56

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The number of wrong attempts = $70n = 56$

Subject T

The number of questions skipped = $(48+60) = 108$

The number of questions attempted = $\{(108/45*55)\} = 132$

The total number of questions contained = $(108+132) = 240$

The number of correct attempts = 75% of $132 = 99$

The number of wrong attempts = 25% of $132 = 33$

Subject	The total number of questions contained	The number of questions attempted	The number of questions skipped	The number of correct attempts	The number of wrong attempts
P	120	90	30	54	36
Q	200	160	40	144	16
R	80	72	8	54	18
S	160	112	48	56	56
T	240	132	108	99	33
Total	800	566	234	407	159

Answer: C

Subject R (another student)

The total number of questions contained = 80

The number of questions attempted = $3/4$ of $80 = 60$

The number of questions skipped = $1/4$ of $80 = 20$

Let the number of correct attempts be k .

So, the number of wrong attempts = $(60-k)$

Therefore, the marks obtained = $[(k*1)-\{(60-k)*0.25\}] = \{k-(15-0.25k)\} = (1.25k-15)$

Now,

$$1.25k - 15 = 55$$

$$1.25k = 70$$

$$k = 56$$

So, the number of correct attempts = $k = 56$

Therefore, the required percentage = $\{(56/60)*100\}\% = 93.33\%$.

Hence, the correct answer is option C.

20. Questions

Let the total number of questions contained in subject P and Q 300m and 500m respectively.

Subject P

The total number of questions contained = 300m

The number of questions attempted = 75% of 300m = 225m

The number of questions skipped = 25% of 300m = 75m

The number of correct attempts = 60% of 225m = 135m

The number of wrong attempts = 40% of 225m = 90m

Subject Q

The total number of questions contained = 500m

The number of questions attempted = 80% of 500m = 400m

The number of questions skipped = 20% of 500m = 100m

The number of correct attempts = 90% of 400m = 360m

The number of wrong attempts = 10% of 400m = 40m

Now,

$$90m - 40m = 20$$

$$50m = 20$$

$$m = 0.40$$

Also, the ratio the of the total number of questions contained in subject R and S = 1:2. Let it be 100n and 200n respectively.

Subject R

The total number of questions contained = 100n

The number of questions attempted = 90% of 100n = 90n

The number of questions skipped = 10% of 100n = 10n

The number of correct attempts = 75% of 90n = 67.50n

The number of wrong attempts = 25% of 90n = 22.50n

Subject S

The total number of questions contained = 200n

The number of questions attempted = 70% of 200n = 140n

The number of questions skipped = 30% of 200n = 60n

The number of correct attempts = 50% of 140n = 70n

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The number of wrong attempts = 50% of 140n = 70n

Now,

$$70n - 67.50n = 2$$

$$2.50n = 2$$

$$n = 0.80$$

Subject P

The total number of questions contained = 300m = (300*0.40) = 120

The number of questions attempted = 225m = (225*0.40) = 90

The number of questions skipped = 75m = (75*0.40) = 30

The number of correct attempts = 135m = (135*0.40) = 54

The number of wrong attempts = 90m = (90*0.40) = 36

Subject Q

The total number of questions contained = 500m = (500*0.40) = 200

The number of questions attempted = 400m = (400*0.40) = 160

The number of questions skipped = 100m = (100*0.40) = 40

The number of correct attempts = 360m = (360*0.40) = 144

The number of wrong attempts = 40m = (40*0.40) = 16

Subject R

The total number of questions contained = 100n = (100*0.80) = 80

The number of questions attempted = 90n = (90*0.80) = 72

The number of questions skipped = 10n = (10*0.80) = 8

The number of correct attempts = 67.50n = (67.50*0.80) = 54

The number of wrong attempts = 22.50n = (22.50*0.80) = 18

Subject S

The total number of questions contained = 200n = (200*0.80) = 160

The number of questions attempted = 140n = (140*0.80) = 112

The number of questions skipped = 60n = (60*0.80) = 48

The number of correct attempts = 70n = (70*0.80) = 56

The number of wrong attempts = 70n = 56

Subject T

The number of questions skipped = (48+60) = 108

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The number of questions attempted = $\{(108/45 \times 55)\} = 132$

The total number of questions contained = $(108+132) = 240$

The number of correct attempts = 75% of $132 = 99$

The number of wrong attempts = 25% of $132 = 33$

Subject	The total number of questions contained	The number of questions attempted	The number of questions skipped	The number of correct attempts	The number of wrong attempts
P	120	90	30	54	36
Q	200	160	40	144	16
R	80	72	8	54	18
S	160	112	48	56	56
T	240	132	108	99	33
Total	800	566	234	407	159

Answer: A

Subject U

The total number of questions contained = 280

The number of questions skipped = $2Y$

The total number of questions attempted = $(280-2Y)$

The number of wrong attempts = $\{(2Y/2) \times 3\} = 3Y$

The number of correct attempts = $\{(280-2Y)-3Y\} = (280-5Y)$

Now,

$(280-5Y)/(280-2Y) = 5/6$ (**Note: $83.33\% = 5/6$**)

$1680 - 30Y = 1400 - 10Y$

$20Y = 280$

$Y = 14$

Now, the total number of questions attempted in subject T = 132

And, the total number of questions attempted in subject U = $(280-2Y) = (280-2 \times 14) = (280-28) = 252$

Therefore, the required difference = $(252-132) = 120$.

Hence, the correct answer is option A.

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